Notes on Complex Variables MTH 5103 2024-2025

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Abstract

What is the use of complex functions and complex variables? One of many things, foe example, to compute the actual value of real convergent series. We know, for example, that the series $\sum_{n=1}^{\infty} \frac{1}{n^2} < \infty$ convergent. The question is $\sum_{n=1}^{\infty} \frac{1}{n^2} =$? Towards the end of this module we will be able to compute the answer. We will see

$$1 + \frac{1}{2^2} + \frac{1}{3^2} + \cdots = \frac{\pi^2}{6}$$
$$1 + \frac{1}{2^4} + \frac{1}{3^4} + \cdots = \frac{\pi^4}{90}$$
$$1 + \frac{1}{2^6} + \frac{1}{3^6} + \cdots = \frac{\pi^6}{245}$$

Using complex functions we will be able to compute quite complicated (*real!*) integrals, which may be very hard to compute otherwise and a lot of other things.

Check: *Compute:* $\int_{-\infty}^{+\infty} \frac{dx}{1+x^4}$

We start with the basic definitions, properties, and the geometry of complex numbers.

Remark: One of the most annoying things about LaTeX, and there are a lot, is how hard it is to just start writing math papers. This is an empty template intended to enable one to just start writing a paper. It uses my personal style, but might be useful for other people. It uses BibLatex by default. Search `DELETE` in the file to know which portion to delete when you start writing.

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Week 1

Lecture 1: Intro and Complex Numbers

1 Introduction

These notes cover the fundamental concepts of complex variables. The course begins with the basics of complex numbers, their algebra, and geometry. We then move to sequences, series, and power series, and finally to the core concepts of analytic functions, Cauchy-Riemann equations, and various related theorems and applications.

1.1 Motivation

What is the use of complex functions and complex variables? One of many things is to compute the actual value of real convergent series. For example, we know that the series $\sum_{n=1}^{\infty} \frac{1}{n^2}$ converges. The question is, what is the value of the sum: $\sum_{n=1}^{\infty} \frac{1}{n^2} =?$ Towards the end of this module, we will be able to compute the answer. We will see:

- $1 + \frac{1}{2^2} + \frac{1}{3^2} + \cdots = \frac{\pi^2}{6}$
- $1 + \frac{1}{2^4} + \frac{1}{3^4} + \cdots = \frac{\pi^4}{90}$
- $1 + \frac{1}{2^6} + \frac{1}{3^6} + \cdots = \frac{\pi^6}{945}$

Using complex functions we will be able to compute quite complicated (real!) integrals, which may be very hard to compute otherwise, and a lot of other things. For example: Compute $\int_{-\infty}^{\infty} \frac{1}{x^4+1} dx =?$

2 The Field of Complex Numbers

We start with the basic definitions, properties, and the geometry of complex numbers.

2.1 Basic definitions and algebra

Definition 2.1. A *complex number* is an ordered pair of real numbers, (x, y). If $y = 0$, we get the pair $(x, 0) = x$ (which we denote simply by x).

Since real numbers are a part of complex numbers, the algebraic operations (sum, subtraction, multiplication, division) on complex numbers need to be such that they agree with the corresponding algebraic operations on real numbers. We denote $\mathbb{C}$ as the set of all complex numbers.

Definition 2.2. Two complex numbers $z_1 = (x_1, y_1)$ and $z_2 = (x_2, y_2)$, $z_1, z_2 \in \mathbb{C}$, are *equal* if and only if $x_1 = x_2$ and $y_1 = y_2$.

Definition 2.3. The *sum* of two complex numbers $z_1 = (x_1, y_1)$ and $z_2 = (x_2, y_2)$ is a complex number $z_1 + z_2 = (x_1 + x_2, y_1 + y_2) = z_3 \in \mathbb{C}$.

Definition 2.4. The *product* of two complex numbers $z_1 = (x_1, y_1)$ and $z_2 = (x_2, y_2)$ is a complex number $z_1 z_2 = (x_1 x_2 - y_1 y_2, x_1 y_2 + x_2 y_1) = z_3 \in \mathbb{C}$.

It is easy to see that if we apply this rule to real numbers, we get the usual product. For $x_1, x_2 \in \mathbb{R}$: $x_1 = (x_1, 0)$, $x_2 = (x_2, 0)$

$$(x_1, 0)(x_2, 0) = (x_1 x_2 - 0 \cdot 0, x_1 \cdot 0 + 0 \cdot x_1) = (x_1 x_2, 0) = x_1 x_2.$$

Usually, we will write a complex number as $z = x + iy$, where $i = (0, 1)$. i is an ordered pair (0,1). Let us compute i^2 using definition 1.4:

$$i^2 = (0, 1)(0, 1) = (0 \cdot 0 - 1 \cdot 1, 0 \cdot 1 + 1 \cdot 0) = (-1, 0) = -1.$$

It is easy to check the usual rules for arithmetic, namely the axioms of a field.

Proposition 2.5

If $z_1, z_2, z_3 \in \mathbb{C}$, then:

1. **Closure:** $z_1 + z_2 \in \mathbb{C}$, $z_1 z_2 \in \mathbb{C}$.
2. **Commutativity:** $z_1 + z_2 = z_2 + z_1$, $z_1 z_2 = z_2 z_1$.
3. **Associativity:** $z_1 + (z_2 + z_3) = (z_1 + z_2) + z_3$, $z_1(z_2 z_3) = (z_1 z_2) z_3$.
4. **Distributivity:** $z_1(z_2 + z_3) = z_1 z_2 + z_1 z_3$.
5. **Zero and Identity:** $z + 0 = z$, $z \cdot 1 = z$.
6. **Negative and Inverse:** $z + (-z) = 0$, where for $z = x + iy$, $-z = -x - iy$. If $z \neq 0$, then $z \cdot z^{-1} = 1$, where $z^{-1} = \frac{x}{x^2+y^2} - i\frac{y}{x^2+y^2}$.

Proof. Let us prove the Inverse. As for the rest: (**Check!**) - check at home (all parts follow from the analogous properties of real numbers).

$$z \cdot z^{-1} = (x + iy)\left(\frac{x}{x^2+y^2} - i\frac{y}{x^2+y^2}\right) = \frac{x^2 - ixy + ixy - i^2 y^2}{x^2+y^2} + i \cdot 0 = \frac{x^2+y^2}{x^2+y^2} + i \cdot 0 = 1. \qquad \square$$

We will see why the inverse is defined in exactly this way.

Definition 2.6. Let $z = x + iy \in \mathbb{C}$. x is called the *real part* of z and denoted by $x = \Re(z)$; y is called the *imaginary part* of z, denoted by $y = \Im(z)$.

The algebraic form $x+iy$ to write complex numbers allows us to add and multiply complex numbers as polynomials, but remember that $i^2 = -1$. For example:

$$(x_1 + iy_1)(x_2 + iy_2) = x_1x_2 + ix_1y_2 + iy_1x_2 + i^2y_1y_2 = (x_1x_2 - y_1y_2) + i(x_1y_2 + y_1x_2).$$

Lecture 2: more on Complex Numbers.

Example 2.7

Let $z_1 = 1 + i$, $z_2 = 2 - i$. Compute $z_1 + z_2$, z_1z_2, z_2^{-1}.

Solution. $z_1 + z_2 = (1 + i) + (2 - i) = (1 + 2) + (1 + (-1))i = 3$ $z_1z_2 = (1 + i)(2 - i) = (1 \cdot 2 + i(-i)) + i(1(-1) + 1 \cdot 2) = 3 + i$ z_2^{-1}: $\Re(z_2) = x = 2$, $\Im(z_2) = y = -1 \implies z_2^{-1} = \frac{x}{x^2+y^2} - i\frac{y}{x^2+y^2} = \frac{2}{5} + i\frac{1}{5}$ □

2.2 Complex conjugate and division

Definition 2.8. The *complex conjugate* of the complex number $z = x + iy$ is the complex number given by $\bar{z} = x - iy$ (the bar denotes the operation of complex conjugation).

Remark 2.9. Check: On your own mate.

1. $\overline{(\bar{z})} = z$
2. $\overline{z_1 + z_2} = \bar{z_1} + \bar{z_2}$
3. $\overline{z_1z_2} = \bar{z_1}\bar{z_2}$
4. $\Re(z) = \frac{1}{2}(z + \bar{z}) = \frac{1}{2}(x + iy + x - iy) = x$
5. $\Im(z) = \frac{1}{2i}(z - \bar{z}) = \frac{1}{2i}(x + iy - x + iy) = y$

Using addition and multiplication we can define subtraction and division.

Definition 2.10. Subtraction: $z_1 - z_2$ is a complex number z_3 such that $z_1 = z_2 + z_3$.

Using this definition we obtain: $z_3 = (x_1 - x_2) + i(y_1 - y_2)$

Definition 2.11. Division: Let $z_2 \neq 0$. $\frac{z_1}{z_2}$ is a complex number z_3 such that $z_1 = z_2z_3$.

Let us calculate z_3. From the definition for $z_1 = x_1 + iy_1$, $z_2 = x_2 + iy_2$, $z_3 = x_3 + iy_3$ we get: $z_1 = x_1 + iy_1 = (x_2 + iy_2)(x_3 + iy_3) = (x_2x_3 - y_2y_3) + i(y_2x_3 + x_2y_3) = z_2z_3$ Using the equality definition we have the following system of equations:

$$\begin{cases} \Re(z_1) = x_1 = x_2x_3 - y_2y_3 = \Re(z_2z_3) \\ \Im(z_1) = y_1 = y_2x_3 + x_2y_3 = \Im(z_2z_3) \end{cases}$$

Let us solve this system: the unknowns are x_3, y_3. Solving it gives:

$$x_3 = \frac{x_1x_2 + y_1y_2}{x_2^2 + y_2^2}, \quad y_3 = \frac{x_2y_1 - x_1y_2}{x_2^2 + y_2^2}$$

Therefore,

$$z_3 = \frac{z_1}{z_2} = \frac{x_1x_2 + y_1y_2}{x_2^2 + y_2^2} + i\frac{x_2y_1 - x_1y_2}{x_2^2 + y_2^2}$$

In particular, we obtain for $1/z$: $x_1 = 1, y_1 = 0, x_2 = x, y_2 = y$,

$$\frac{1}{z} = \frac{1 \cdot x + 0 \cdot y}{x^2 + y^2} + i\frac{x \cdot 0 - 1 \cdot y}{x^2 + y^2} = \frac{x}{x^2 + y^2} - i\frac{y}{x^2 + y^2}$$

Note that in the denominator we have $z_2\bar{z}_2 = (x_2 + iy_2)(x_2 - iy_2) = x_2^2 + y_2^2$.

Therefore, the recipe for division is: multiply the numerator and the denominator by the complex conjugate of the denominator and separate the real and imaginary parts.

Example 2.12

Calculate $z = \frac{3+4i}{1-i}$.

Solution.

$$z = \frac{3+4i}{1-i} = \frac{3+4i}{1-i} \cdot \frac{1+i}{1+i} = \frac{3+3i+4i+4i^2}{1^2-(-1)^2} = \frac{-1+7i}{2} = -\frac{1}{2} + i\frac{7}{2}$$

□

Example 2.13

Prove that for any $z \in \mathbb{C}$, $\frac{z+\bar{z}}{z\bar{z}} \in \mathbb{R}$.

Solution. $\frac{z+\bar{z}}{z\bar{z}} = \frac{z}{z\bar{z}} + \frac{\bar{z}}{z\bar{z}} = \frac{1}{\bar{z}} + \frac{1}{z} = \frac{z+\bar{z}}{z\bar{z}} = \frac{2\Re z}{|z|^2} = \frac{2x}{x^2+y^2} \in \mathbb{R}$. □

Definition 2.14. The *modulus* of $z = x + iy$ is $|z| = \sqrt{x^2 + y^2}$.

Remark 2.15. $|z| \geq 0$ always and $|z| = 0 \iff z = 0 \iff x = y = 0$.

Check: (Simple relation) $|z| \geq \Re(z)$ and $|z| \geq \Im(z)$. Observe: $|z| = \sqrt{x^2 + y^2} \implies |z|^2 = z\bar{z} \implies \frac{1}{z} = \frac{\bar{z}}{|z|^2}$ if $z \neq 0$.

In the following proposition we list and prove the properties of the modulus of a complex number.

Proposition 2.16

Let $z, w \in \mathbb{C}$. Then

1. $|zw| = |z||w|$
2. $|z+w| \leq |z| + |w|$ (Triangle Inequality)
3. $|z-w| \geq ||z| - |w||$ (Reverse Triangle inequality)

Proof. 1. Observe that (commutativity): $|zw|^2 = (zw)(\overline{zw}) = (zw)(\bar{z}\bar{w}) = (z\bar{z})(w\bar{w}) = |z|^2|w|^2$ (for any z_1, z_2: $\overline{z_1 z_2} = \bar{z}_1\bar{z}_2$). This proves 1) after taking square roots.

2. By the definition: For any z_1, z_2: $\overline{z_1 + z_2} = \bar{z}_1 + \bar{z}_2$.

$$\begin{aligned}
|z+w|^2 &= (z+w)(\bar{z}+\bar{w}) = (z+w)(\overline{z+w}) = z\bar{z} + w\bar{w} + z\bar{w} + \bar{z}w \\
&= |z|^2 + |w|^2 + z\bar{w} + \overline{(z\bar{w})} = |z|^2 + |w|^2 + 2\Re(z\bar{w}) \\
&\boxed{\forall z, w : w\bar{w} = \bar{w}w,\ z\bar{z} = \bar{z}z; \quad z + \bar{z} = 2\Re(z)} \\
&\leq |z|^2 + |w|^2 + 2|z\bar{w}| = |z|^2 + |w|^2 + 2|z||w| = |z|^2 + |w|^2 + 2|z||w| \\
&= (|z| + |w|)^2. \\
&\boxed{(|z_1 z_2| = |z_1||z_2| \forall z_1, z_2).}
\end{aligned}$$

3. We employ the trick of adding zero and apply 2).

$$|z| = |(z-w) + w| \leq |z-w| + |w| \implies |z-w| \geq |z| - |w| \tag{4}$$

Similarly:

$$|w-z| = |-(z-w)| = |z-w| \geq |w| - |z| \tag{5}$$

Combining 4 and 5 we get 3). □

Example 2.17

Prove that if $|z_1| = |z_2| = 1$, then $\frac{z_1+z_2}{1+z_1z_2} \in \mathbb{R}$.

Solution. First multiply and divide by the complex conjugate of the denominator. Denominator $1 + z_1z_2 = \overline{1 + z_1z_2} = 1 + \bar{z}_1\bar{z}_2$

$$\begin{aligned}
\frac{z_1+z_2}{1+z_1z_2} &= \frac{z_1+z_2}{1+z_1z_2} \cdot \frac{1+\bar{z}_1\bar{z}_2}{1+\bar{z}_1\bar{z}_2} \\
&= \frac{z_1 + z_2 + z_1\bar{z}_1\bar{z}_2 + z_2\bar{z}_1\bar{z}_2}{1 + z_1z_2 + \bar{z}_1\bar{z}_2 + z_1z_2\bar{z}_1\bar{z}_2}
\end{aligned}$$

$$= \frac{(z_1 + \bar{z}_1) + (z_2 + \bar{z}_2)}{|1 + z_1 z_2|^2} (|z_1|{=}|z_2|{=}\, 1)$$
$$= \frac{2\Re(z_1) + 2\Re(z_2)}{|1 + z_1 z_2|^2} \in \mathbb{R}$$

□

Example 2.18

Prove that $\frac{z-1}{z+1}$ is purely imaginary iff $|z| = 1$.

Solution. (Purely imaginary, namely its real part is equal to 0) Denominator $z+1 = \bar{z}+1$

$$\frac{z-1}{z+1} = \frac{z-1}{z+1} \cdot \frac{\bar{z}+1}{\bar{z}+1} = \frac{z\bar{z} - z + \bar{z} - 1}{|z+1|^2} = \frac{|z|^2 - 1 + (z - \bar{z})}{|z+1|^2} = \frac{|z|^2 - 1 + 2i\Im(z)}{|z+1|^2}$$

Note that $|z|^2 \in \mathbb{R}$, $|z+1|^2 \in \mathbb{R}$, $\Im(z) \in \mathbb{R}$.

$$\Re(\frac{z-1}{z+1}) = \frac{|z|^2 - 1}{|z+1|^2} \quad \Im(\frac{z-1}{z+1}) = \frac{2\Im(z)}{|z+1|^2}$$

$$\Re(\frac{z-1}{z+1}) = 0 \iff \frac{|z|^2 - 1}{|z+1|^2} = 0 \iff |z|^2 - 1 = 0 \iff |z|^2 = 1 \iff |z| = 1$$

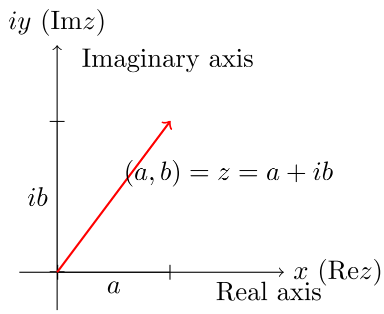
□

Lecture 2+3: Geometry in Complex Numbers. WO00o0W.

3 Geometric Representation of Complex Numbers

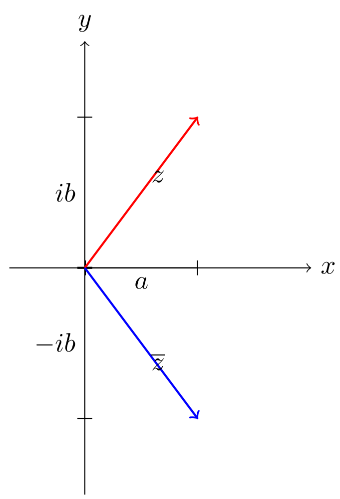
3.1 Modulus and Argument. Polar Form

Any complex number $z = a + ib$ can be numerically described as a point in the plane with the coordinates (a, b) or as a vector that starts at the origin $(0, 0)$ and ends at the point (a, b). iy $\Im z$ Imaginary axis ib Point in the plane $\mathbb{R}^2$: (a, b). $(a, b) = z = a + ib$ OR vector connecting $(0, 0)$ and (a, b).



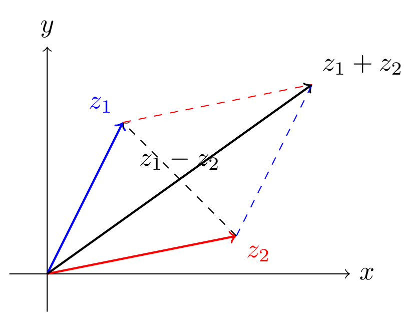
x ($\Re z$) Real Axis

The plane (x, y) is called the complex plane. The x-axis is called the real axis and the y-axis is called the imaginary axis.

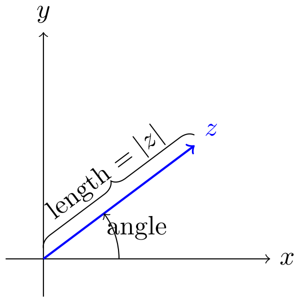


The complex conjugate of z in this plane is described by a vector that is symmetric with respect to the real axis.

The sum of 2 complex numbers can be described as a sum of two vectors that start at the origin: the diagonal of the parallelogram that is constructed out of those 2 vectors.



The difference between 2 complex numbers is the second(shorter) diagonal of the same

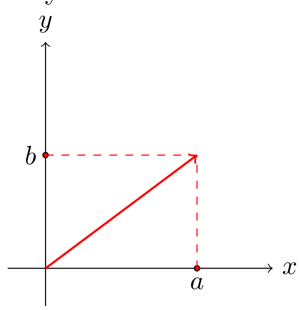


parallelogram.

Recall: any vector is uniquely determined by its length and the angle that it creates with the x-axis [in the counterclockwise, positive direction]

Definition 3.1 (Alternative)**.** The distance from the origin $(0, 0)$ to the point $z = (x, y)$ is called the modulus of the complex number z.

Let us assume that this is the only definition and obtain previous definition from it. By



the Pythagorean theorem we get:

$|z| = \sqrt{a^2 + b^2}(= |a + ib|)$. This is exactly the previous modulus definition.

Example 3.2

Compute $|z|$ for $z_1 = 7 + 9i$, $z_2 = 4 - 3i$. $|z_1| = |7 + 9i| = \sqrt{7^2 + 9^2} = \sqrt{130}$ $|z_2| = |4 - 3i| = \sqrt{4^2 + (-3)^2} = \sqrt{25} = 5$

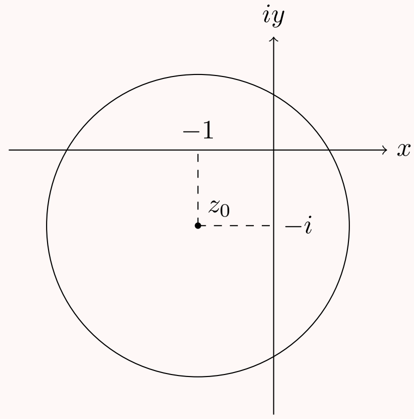
Check (very easy!): The distance between any two points $z = (x, y)$ and $z_0 = (x_0, y_0)$ is

$$|z - z_0| = |(x - x_0) + i(y - y_0)| = \sqrt{(x - x_0)^2 + (y - y_0)^2}$$

The equations $|z_1| = \sqrt{130}$, $|z_2| = 5$ describe the geometric place of all the points z that are at distance $\sqrt{130}$ or 5 from the origin - namely, it is a circle. In general we have: Circle with radius R centered at z_0: $|z - z_0| = R$.

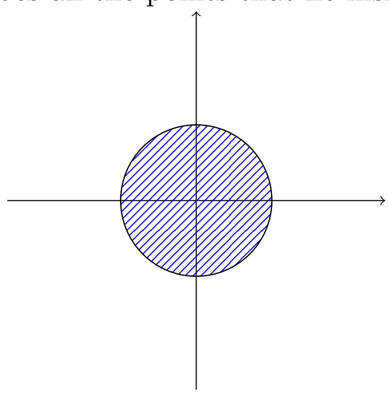
Example 3.3

$|z + 1 + i| = 2$: circle centered at $z_0 = -(1 + i)$ of radius 2.



Indeed, all the points for which this equation holds lie on the circle described by $\sqrt{(x+1)^2+(y+1)^2}=2$

The equation $|z-z_0|<R$ describes all the points that lie inside the circle centered at z_0

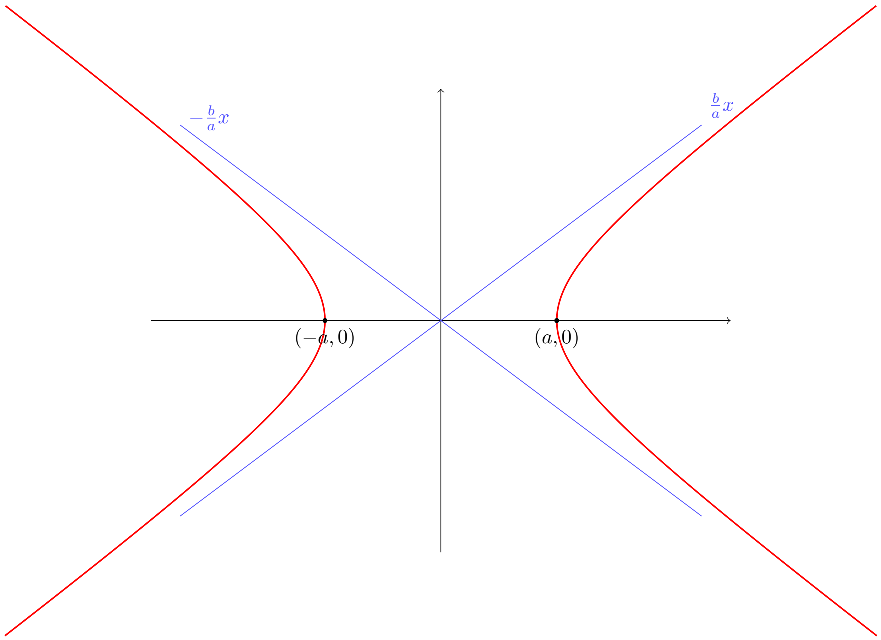


of radius R (but not on the circle!).

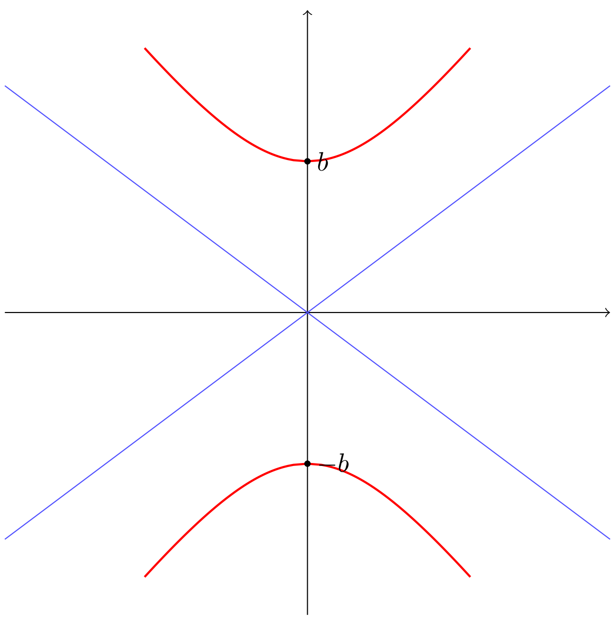
Example 3.4

Find the geometric place of the points z for which $|z+2|-|z-2|=1$

We are looking for the geometric place of all points z such that the difference of distances from the point z to two points $z=2$ and $z=-2$ is 1. Let us show that this is a hyperbola. Recall: hyperbola equation $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ - hyperbola with vertices at $(a,0)$ and $(-a,0)$ with asymptotes at $y=\pm\frac{b}{a}x$.



For $-\frac{y^2}{b^2}+\frac{x^2}{a^2}=1$: the vertices are at $(0,b)$ and $(0,-b)$ and asymptotes at $y=\pm\frac{b}{a}x$.

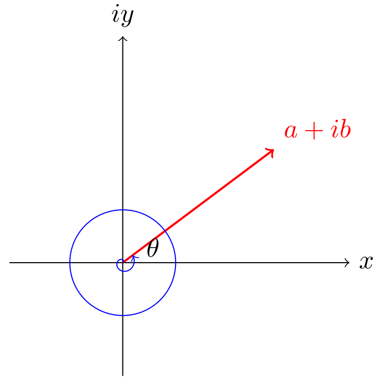


Here we have: $|z+2|-|z-2|=1 \implies |x+iy+2|-|x+iy-2|=1 \implies |(x+2)+iy|-|(x-2)+iy|=1 \implies \sqrt{(x+2)^2+y^2}-\sqrt{(x-2)^2+y^2}=1 \implies \sqrt{(x+2)^2+y^2}=1+\sqrt{(x-2)^2+y^2} \iff (x+2)^2+y^2=1+(x-2)^2+y^2+2\sqrt{(x-2)^2+y^2} \iff (x+2)^2-(x-2)^2-1=2\sqrt{(x-2)^2+y^2} \iff x^2+4x+4-x^2+4x-4-1=2\sqrt{(x-2)^2+y^2} \iff 8x-1=2\sqrt{(x-2)^2+y^2} \iff (8x-1)^2=4((x-2)^2+y^2) \iff 64x^2-16x+1=4x^2-16x+16+4y^2 \implies 60x^2-4y^2=15 \iff 4x^2-\frac{y^2}{15/4}=1 \iff \frac{x^2}{1/4}-\frac{y^2}{15/4}=1 \iff \frac{x^2}{(1/2)^2}-\frac{y^2}{(\sqrt{15}/2)^2}=1$ Thus, this is a hyperbola with vertices at $(\pm\frac{1}{2},0)$ with asymptotes $y=\pm\frac{\sqrt{15}/2}{1/2}x=\pm\sqrt{15}x$

Now we define the second object needed for a unique definition of a vector - an angle. This angle is called the argument of a complex number.

Definition 3.5. An *argument* of $z \neq 0$, denoted by $\arg z$, is an angle θ such that $\cos\theta = \frac{\Re z}{|z|}$, $\sin\theta = \frac{\Im z}{|z|}$.

It is only defined up to the addition of multiples of 2π. We say that θ is the *principle value* of the argument if $0 \leq \theta < 2\pi$ and denote the principal value of the argument z by Arg$z$.



The argument is exactly the angle that the vector z creates with the x-axis when measured following the positive (anti-clockwise) direction. As we said: the argument takes infinitely many values that differ one from another by a multiple of 2π. So, now we have 3 ways to represent a complex number: as an ordered pair, an algebraic form $x+iy$, or via its modulus and an argument. Representation: $z=(x,y), z=x+iy$, or $x=|z|\cos\theta, y=|z|\sin\theta \implies z=|z|(\cos\theta+i\sin\theta)$ Polar form (r,θ - coordinates.) The last form is called the polar form of a complex number.

It is more convenient to use the algebraic form to add complex numbers and it is easier to multiply using the polar form. Let us see how we pass from an algebraic to polar representation.

Example 3.6

Write the following numbers in polar form:

(a) $-2+2i$

(b) 2

(c) $-i$

Solution. a) $z=-2+2i \implies |z|=\sqrt{(-2)^2+2^2}=2\sqrt{2}$, $\cos\theta=\frac{-2}{2\sqrt{2}}=-\frac{1}{\sqrt{2}}$, $\sin\theta=\frac{2}{2\sqrt{2}}=\frac{1}{\sqrt{2}} \implies \theta=\frac{3\pi}{4}(+2\pi k, k\in\mathbb{Z}) \implies z=2\sqrt{2}(\cos\frac{3\pi}{4}+i\sin\frac{3\pi}{4})$

b) $z = 2 \implies |z| = \sqrt{2^2 + 0^2} = 2$, $\cos\theta = \frac{2}{2} = 1$, $\sin\theta = \frac{0}{2} = 0 \implies \theta = 0(+2\pi k, k \in \mathbb{Z})$, $z = 2(\cos 0 + i \sin 0)$

c) $z = -i \implies |z| = \sqrt{0^2 + (-1)^2} = 1$, $\cos\theta = \frac{0}{1} = 0$, $\sin\theta = \frac{-1}{1} = -1 \implies \theta = \frac{3\pi}{2}(+2\pi k, k \in \mathbb{Z}) \implies z = 1(\cos(\frac{3\pi}{2}) + i \sin(\frac{3\pi}{2}))$

□

In all previous examples the argument is up to the addition of $2\pi k$.

Remark 3.7. $a \equiv b(\mathrm{mod}\ c)$: means that there exists an integer $k \in \mathbb{Z}$ s.t. $a - b = kc$. In particular, if $c = 2\pi$, then $a \equiv b(\mathrm{mod}\ 2\pi)$ means that there exists $k \in \mathbb{Z}$ s.t. $a - b = 2\pi k$. **Important**: The argument of $z = 0$ is not defined!

We have already studied some of the properties of the modulus in proposition 2.16.

Proposition 3.8

For any $z_1, z_2 \in \mathbb{C}(z_2 \neq 0)$.

1. $|\bar{z}| = |z|$
2. $z\bar{z} = |z|^2$
3. $|\frac{z_1}{z_2}| = \frac{|z_1|}{|z_2|}$

Proof. Here we present the proof of 1) via polar coordinates; **Check!**the rest including the ones from proposition 2.16 (in the same way). Let $z = |z|(\cos\theta + i \sin\theta)$. $\bar{z} = |z|(\cos\theta - i \sin\theta)$ $|z| = |z|\sqrt{\cos^2\theta + \sin^2\theta} = |z|$ $|\bar{z}| = |z|\sqrt{\cos^2\theta + (-\sin\theta)^2} = |z|\sqrt{\cos^2\theta + \sin^2\theta} = |z|$ □

Let us study the properties of the argument.

Proposition 3.9

For any $z_1, z_2 \in \mathbb{C}(z_1, z_2 \neq 0)$

1. $\arg(z_1 z_2) = \arg z_1 + \arg z_2$
2. $\arg(\frac{z_1}{z_2}) = \arg z_1 - \arg z_2$

Here we prove 1). Prove 2) in the same way.

Proof. Write z_1, z_2 in polar form: let $r_1 = |z_1|$, $r_2 = |z_2|$, $\phi_1 = \arg z_1$, $\phi_2 = \arg z_2$. Then $z_1 = r_1(\cos\phi_1 + i \sin\phi_1)$,$z_2 = r_2(\cos\phi_2 + i \sin\phi_2)$

$z_1/z_2 = r_1/r_2(\cos\phi_1/\cos\phi_2 + i \sin\phi_1/\sin\phi_2)$ (4) $z_1 z_2 = r_1 r_2(\cos\phi_1\cos\phi_2 - \sin\phi_1\sin\phi_2 + i(\sin\phi_1\cos\phi_2 + \sin\phi_2\cos\phi_1)) = r_1 r_2 = r_1 r_2(\cos(\phi_1 + \phi_2) + i \sin(\phi_1 + \phi_2))$

Since $\arg z_1 = \phi_1 + 2\pi k$, $\arg z_2 = \phi_2 + 2\pi m \implies \arg(z_1 z_2) = \phi_1 + \phi_2 + 2\pi n$, where $n = m + k$ (1)

$\square$

Exercise 3.10. 1. $\implies$ The Geometric rule for multiplication: multiply moduli, add arguments.

From rule: If $z_1 = z_2 = z$, from (4): $z^2 = r^2(\cos 2\phi + i \sin 2\phi)$. It is easy to prove by induction on n: **De Moivre formula** $z^n = |z|^n(\cos(n\phi) + i \sin(n\phi))$

Example 3.11

Express $\cos 3\alpha$ and $\sin 3\alpha$ in terms of $\cos\alpha$, $\sin\alpha$.

Solution. Construct complex number $z = \cos\alpha + i\sin\alpha$. By De Moivre formula $z^3 = \cos 3\alpha + i \sin 3\alpha$. On the other hand: $z^3 = (\cos\alpha + i\sin\alpha)^3 = \cos^3\alpha + 3i\cos^2\alpha\sin\alpha + 3i^2\cos\alpha\sin^2\alpha + i^3\sin^3\alpha = \cos^3\alpha - 3\cos\alpha\sin^2\alpha + i(3\cos^2\alpha\sin\alpha - \sin^3\alpha)$

By the equality definition: $z_1 = z_2 \iff \Re(z_1) = \Re(z_2)$ and $\Im(z_1) = \Im(z_2)$. Let us compare these two expressions: $\cos 3\alpha + i\sin 3\alpha = \cos^3\alpha - 3\cos\alpha\sin^2\alpha + i(3\cos^2\alpha\sin\alpha - \sin^3\alpha) \implies \cos 3\alpha = \cos^3\alpha - 3\cos\alpha\sin^2\alpha$ $\sin 3\alpha = 3\cos^2\alpha\sin\alpha - \sin^3\alpha$ $\square$

Example 3.12

Define $\frac{1}{z}$ geometrically.

Solution. Let us compute the modulus and the argument. By proposition 3.8 part 3): $|\frac{1}{z}| = \frac{1}{|z|}$ and since $z \cdot \frac{1}{z} = 1$ $0 = \arg 1 = \arg(z \cdot \frac{1}{z}) = \arg z + \arg\frac{1}{z} \implies \arg\frac{1}{z} = -\arg z$ $\square$

3.2 Roots and Euler's formula

Definition 3.13. n-th root of a complex number z is a complex number w such that $z = w^n$ (Notation: $\sqrt[n]{z} = w$)

Example 3.14

Find the formula for computing roots of a given $z \in \mathbb{C}$.

Solution. Let $z = w^n$, let us write z and w in polar form. $z = r(\cos\psi + i\sin\psi)$ $w = \rho(\cos\theta + i\sin\theta)$

Now we express ρ and θ in terms of r and ψ (r, ψ are given!). From the definition: $z = w^n$ or $r(\cos\psi + i\sin\psi) = \rho^n(\cos(n\theta) + i\sin(n\theta)) \implies \rho^n = r$, $\cos(n\theta) = \cos\psi$, $\sin(n\theta) = \sin\psi \implies \rho = \sqrt[n]{r}$, $\theta = \frac{\psi + 2\pi k}{n}$ for $0 \le k \le n-1$. It is easy to see (check!)

that for $k > n-1$ we will get again the same values!. Thus, we obtain

$$w = \sqrt[n]{r}(\cos(\frac{\psi + 2\pi k}{n}) + i\sin(\frac{\psi + 2\pi k}{n})), \quad k = 0, 1, 2, \ldots, n-1. \tag{**}$$

□

Corollary 3.15

n-th root of a complex number takes exactly n different values that are placed on the vertices of a regular polygon with n sides inscribed in a circle centered at 0 of radius $\sqrt[n]{|z|}$.

Recall: regular polygon is a polygon whose sides and angles are equal.

Example 3.16

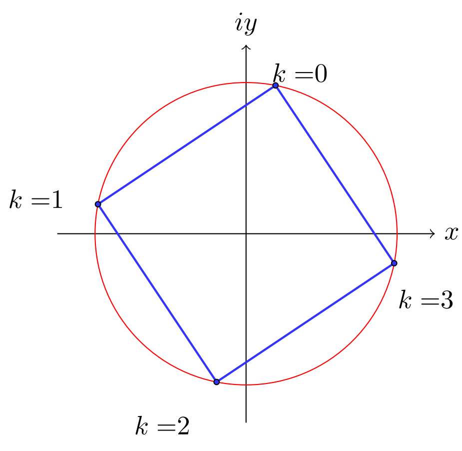
Compute $\sqrt[4]{1-i}$.

Solution. We need to compute the modulus and the argument of 1-i. $|1-i| = \sqrt{1^2 + (-1)^2} = \sqrt{2}$, $\cos\theta = \frac{1}{\sqrt{2}}$, $\sin\theta = \frac{-1}{\sqrt{2}} \implies \theta = \frac{7\pi}{4} + 2\pi k$ (or $\theta = -\frac{\pi}{4} + 2\pi k$) $\implies 1 - i = \sqrt{2}(\cos\frac{7\pi}{4} + i\sin\frac{7\pi}{4})$ - the polar form of 1-i.

Using formula (**) we get $w = \sqrt[n]{1-i} = 2^{1/8}(\cos(\frac{7\pi/4+2\pi k}{4}) + i\sin(\frac{7\pi/4+2\pi k}{4}))$, $k = 0, 1, 2, 3$ $k = 0: \quad 2^{1/8}(\cos\frac{7\pi}{16} + i\sin\frac{7\pi}{16}) \approx 0.21 + i1.07$ $k = 1: \quad 2^{1/8}(\cos\frac{15\pi}{16} + i\sin\frac{15\pi}{16}) \approx -1.07 + i0.21$ $k = 2: \quad 2^{1/8}(\cos\frac{23\pi}{16} + i\sin\frac{23\pi}{16}) \approx -0.21 - i1.07$ $k = 3: \quad 2^{1/8}(\cos\frac{31\pi}{16} + i\sin\frac{31\pi}{16}) \approx 1.07 - i0.21$

Let us draw these roots on a circle, centered at 0 of radius $2^{1/8}$. Connecting the neighboring roots we obtain a square.

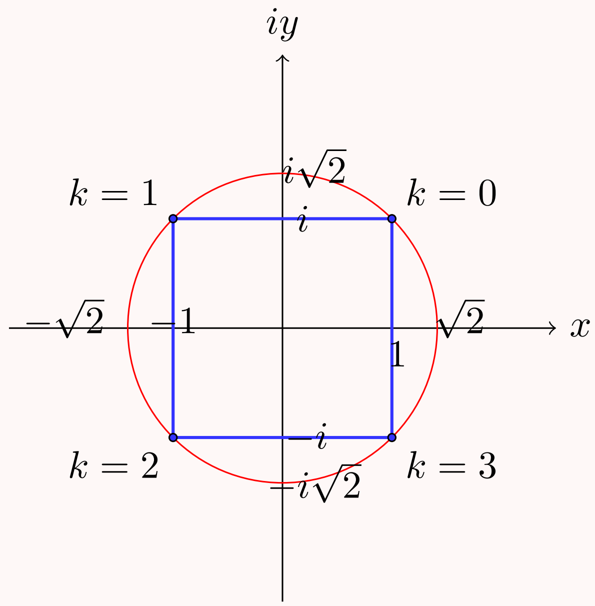
□



Example 3.17

Compute all the roots of $z = \sqrt[4]{-4}$.

Solution. $|-4| = 4$, $\cos\theta = \frac{-4}{4} = -1$, $\sin\theta = \frac{0}{4} = 0 \implies \theta = \pi + 2\pi k$ From (**) we get $\sqrt[4]{-4} = 4^{1/4}(\cos\frac{\pi+2\pi k}{4} + i\sin\frac{\pi+2\pi k}{4}) = \sqrt{2}(\cos(\frac{\pi}{4}+\frac{\pi k}{2}) + i\sin(\frac{\pi}{4}+\frac{\pi k}{2}))$ for $k = 0, 1, 2, 3$.
$k = 0: \quad \sqrt{2}(\cos\frac{\pi}{4} + i\sin\frac{\pi}{4}) = \sqrt{2}(\frac{1}{\sqrt{2}} + i\frac{1}{\sqrt{2}}) = 1 + i$ $k = 1: \quad \sqrt{2}(\cos\frac{3\pi}{4} + i\sin\frac{3\pi}{4}) = \sqrt{2}(-\frac{1}{\sqrt{2}} + i\frac{1}{\sqrt{2}}) = -1 + i$ $k = 2: \quad \sqrt{2}(\cos\frac{5\pi}{4} + i\sin\frac{5\pi}{4}) = \sqrt{2}(-\frac{1}{\sqrt{2}} - i\frac{1}{\sqrt{2}}) = -1 - i$
$k = 3: \quad \sqrt{2}(\cos\frac{7\pi}{4} + i\sin\frac{7\pi}{4}) = \sqrt{2}(\frac{1}{\sqrt{2}} - i\frac{1}{\sqrt{2}}) = 1 - i$



□

The following formula is one of the most celebrated formulas in mathematics.

Theorem 3.18 (Euler's formula; Euler 1760)
Let $\theta \in \mathbb{R}$. Then $\cos\theta + i\sin\theta = e^{i\theta}$ where $e^{i\theta}$ denotes the sum of the power series $e^{i\theta} = 1 + i\theta + \frac{(i\theta)^2}{2!} + \frac{(i\theta)^3}{3!} + ... = \sum_{n=0}^{\infty}\frac{(i\theta)^n}{n!}$.

Set $\theta = \pi$. From this formula: $e^{i\pi} = -1$ - we have connected -1, e, i, π in one formula!

We cannot prove this theorem now: first, we will need to examine the properties of the power series, however, we shall use it as a convenient notation even before we proved it. From the formula: $\Re e^{i\theta} = 1 - \frac{\theta^2}{2!} + \frac{\theta^4}{4!} - ... =$ The Taylor series for $\cos\theta$ (about $\theta = 0$) $\Im e^{i\theta} = \theta - \frac{\theta^3}{3!} + \frac{\theta^5}{5!} - ... =$ The Taylor series for $\sin\theta$ (about $\theta = 0$)

This formula provides a way of conversion between Cartesian coordinates $x + iy$ and the polar form (modulus and argument) as follows: If $z \neq 0$: $|z| = r$, $\arg z = \theta \implies z = re^{i\theta}$ (the polar form)

Then $\bar{z} = x - iy = r(\cos\theta - i\sin\theta) = re^{-i\theta}$. The following should be true: $e^{i\theta} = e^{i(\theta+2\pi n)}$

for every $n \in \mathbb{Z}$ (Euler's formula) $e^{i(\phi+\psi)} = e^{i\phi}e^{i\psi}$ $\frac{1}{e^{i\theta}} = e^{-i\theta}$ $(e^{i\theta})^n = e^{in\theta}$ for every $n \in \mathbb{Z}$.

These facts are easily proved using theorem 3.18, but they are not obvious, since $e^{i\theta}$ means $1 + i\theta + \frac{(i\theta)^2}{2!} + ...$ and raising the real number e to the purely imaginary power $i\theta$ is not well-defined at the moment.

Week 2

Lecture 4: Functions of a Complex Variable

In this lecture we shall discuss the basic properties of functions of one complex variable.

4 Functions of a Complex Variable

4.1 Basic definitions and examples

Definition 4.1. Let $S \subseteq \mathbb{C}$ be a subset of complex numbers. A function f on S is a rule which assigns to each $z \in S$ a complex number (or complex numbers) $f(z)$, called the value of f at z. S is called the Domain of Definition (DoD).

Definition 4.2. If for every z in the DoD corresponds one value w, then we say that f is a single-valued function. If to z correspond several values, then f is a multi-valued function.

Example 4.3

$f(z) = \frac{1}{z}$: f is not defined at $z = 0$ and it is defined and single-valued everywhere else.

For example, the value of f at $z = 8 - 3i$ is

$$f(8-3i) = \frac{1}{8-3i} = \frac{1}{8-3i} \cdot \frac{8+3i}{8+3i} = \frac{8+3i}{8^2+3^2} = \frac{8+3i}{73} = \frac{8}{73} + i\frac{3}{73}$$

Example 4.4

$f(z) = \sqrt{z+1}$ defined everywhere on $\mathbb{C}$ and it is double-valued function: $f(0) = 1$ and $f(0) = -1$ (for instance).

$$f(0) = \sqrt{1} : |1| = 1, \quad \cos\theta = \frac{1}{1} = 1, \quad \sin\theta = \frac{0}{1} = 0 \implies \theta = 0 + 2\pi k$$

$$\implies \sqrt{1} = \cos\left(\frac{0+2\pi k}{2}\right) + i\sin\left(\frac{2\pi k}{2}\right), \quad k = 0, 1$$

$$k = 0 : \cos 0 + i\sin 0 = 1, \quad k = 1 : \cos\pi + i\sin\pi = -1$$

Example 4.5

$f(z) = \sqrt{z-1} + \sqrt{z+1}$ is 4-valued: for example, if $z = 0$, we get $w_1 = i + 1, w_2 = -i + 1, w_3 = i - 1, w_4 = -i - 1$ (**Check!**)

Example 4.6

Polynomials. General form: $P_n(z) = \sum_{j=0}^{n} a_j z^j$ - polynomial of degree n of complex variable z with complex coefficients $a_j \in \mathbb{C}$.

For example: $f(z) = z^2, f(z) = 4 + iz - z^2 + (1+i)z^3$.

Example 4.7

Rational function {ratio of two polynomials}. General form:

$$f(z) = \frac{a_0 + a_1 z + \cdots + a_n z^n}{b_0 + b_1 z + \cdots + b_m z^m}$$

defined and single-valued everywhere, except for the roots (zeros) of the denominator.

Example 4.8

The exponential function $f(z) = e^z$.

Defined to be the series $e^z = \sum_{n=0}^{\infty} \frac{z^n}{n!} = 1 + \frac{z}{1!} + \frac{z^2}{2!} + \ldots$ or equally well by Euler's formula: $e^z = e^x(\cos y + i \sin y)$ {for $z = x + iy$}.

We will see later that the above series converges absolutely for all $z \in \mathbb{C}$, thus $f(z) = e^z$ can be defined for every z in $\mathbb{C}$, namely, we can take the DoD to be $\mathbb{C}$. It is single-valued.

4.2 Real and polar representations

Example 4.9

Trigonometric and Hyperbolic functions. For $z \in \mathbb{C}$ define:

$$\cos z = \frac{1}{2}(e^{iz} + e^{-iz}), \quad \sin z = \frac{1}{2i}(e^{iz} - e^{-iz}) \tag{*}$$

which agrees with the definition for real z (by Euler's formula). By analogy with the real case, we define

$$\cosh z = \frac{1}{2}(e^{z} + e^{-z}), \quad \sinh z = \frac{1}{2}(e^{z} - e^{-z}) \tag{**}$$

Using (*) and (**) it is easy to check that **Check:** $\cos z + i \sin z = e^{iz} = \cosh(iz) + \sinh(iz)$.

Let $z = x + iy$. We rewrite $f(z)$ and separate its real and imaginary parts as follows: (1) $f(z) = \text{Re } f(z) + i\text{Im } f(z) = u(x,y) + iv(x,y)$.

In this way we can think of f as of a real mapping: $f : \mathbb{R}^2 \to \mathbb{R}^2$: (2) $(x,y) \mapsto (u(x,y), v(x,y)) \quad u, v : \mathbb{R}^2 \to \mathbb{R}$.

Let $z = r(\cos\theta + i\sin\theta)$ {the polar form}. Then, in a similar way $f : \mathbb{R}^2 \to \mathbb{R}^2$: (3) $f(r,\theta) = \phi(r,\theta) + i\psi(r,\theta)$.

Example 4.10

Write in the form (1) or (3)

a) $f(z) = z^2$

b) $f(z) = |z|$

c) $f(z) = \text{Re } z$

d) $f(z) = z^n$

e) $f(z) = e^z$

Solution. a) Denote $z = x + iy$. Then $z^2 = (x+iy)^2 = (x^2 - y^2) + i2xy \implies \text{Re } z^2 = x^2 - y^2, \text{Im } z^2 = 2xy$, and $f(x,y) = (x^2 - y^2, 2xy)$.

OR: $z = r(\cos\theta + i\sin\theta) \implies z^2 = r^2(\cos 2\theta + i\sin 2\theta)$, where $r = |z|$, thus $\text{Re } z^2 = r^2\cos 2\theta, \text{Im } z^2 = r^2 \sin 2\theta$, and $f(r,\theta) = (r^2\cos 2\theta, r^2\sin 2\theta)$.

b) $|z| = |x+iy| = \sqrt{x^2+y^2} \implies f(x,y) = (\sqrt{x^2+y^2}, 0) = \sqrt{x^2+y^2} \to$ real function!

c) $z = x+iy, \text{Re } z = x \implies f(x,y) = (x,0) = x \to$ real function! $z = x+iy, \text{Im } z = y \implies$ for $f(z) = \text{Im } z : f(x,y) = (y,0) = y \to$ real!

d) Let $z = r(\cos\theta + i\sin\theta)$ $\{r = |z|\}$. Then, by De Moivre formula $z^n = r^n(\cos(n\theta) + i\sin(n\theta))$, and $f(r,\theta) = (r^n\cos(n\theta), r^n\sin(n\theta))$.

e) Let $z = x+iy$. Then, by Euler's formula: $e^z = e^x(\cos y + i\sin y) \implies f(x,y) = (e^x\cos y, e^x \sin y)$.

□

We can also do the opposite: given a function of x, y we can rewrite it in terms of z and its conjugate: Recall: $\text{Re } z = \frac{1}{2}(z+\bar{z}), \text{Im } z = \frac{1}{2i}(z-\bar{z})$.

Example 4.11

Rewrite in terms z and $\bar{z}$:

a) $f(x,y) = x^2 - y + i(x+y^2)$

b) $f(x,y) = x^2 - 1$

Solution. a)

$$
\begin{aligned}
x^2 - y + i(x+y^2) &= \left(\frac{z+\bar z}{2}\right)^2 - \frac{z-\bar z}{2i} + i\left(\frac{z+\bar z}{2} + \left(\frac{z-\bar z}{2i}\right)^2\right) \\
&= \frac{1}{4}(z^2+2z\bar z+\bar z^2) - \frac{1}{2i}(z-\bar z) + i\left(\frac{1}{2}(z+\bar z) + \frac{1}{4i^2}(z^2-2z\bar z+\bar z^2)\right) \\
&= \frac{1}{4}(z^2+2z\bar z+\bar z^2) - \frac{1}{2i}(z-\bar z) + \frac{i}{2}(z+\bar z) - \frac{i}{4}(z^2-2z\bar z+\bar z^2) \\
&= \frac{1}{4}(z^2+2z\bar z+\bar z^2) + \frac{i}{2}z - \frac{i}{2}\bar z + \frac{i}{2}z + \frac{i}{2}\bar z - \frac{i}{4}(z^2-2z\bar z+\bar z^2) \\
&= \frac{1}{4}(z^2+\bar z^2)(1-i) + \frac{1}{2}z\bar z(1+i) + iz
\end{aligned}
$$

b) $x^2 - 1 = \frac{1}{4}(z^2+\bar z^2+2z\bar z) - 1 = \frac{1}{4}(z^2+\bar z^2+2z\bar z-4)$.

□

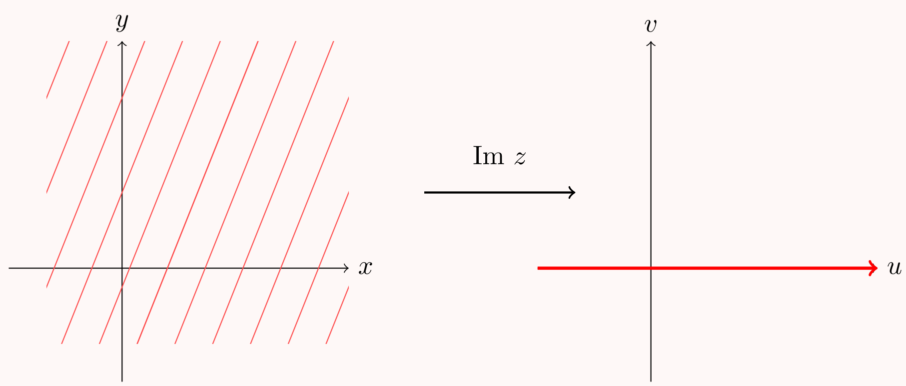
4.3 Transformations of the Complex plane

A complex function is a map (transformation) from $\mathbb{C}$ to $\mathbb{C}$, but to draw a graph of such mapping we would need $2+2=4$ real dimensions.

Instead, we shall examine what happens to various shapes - lines, circles, etc, under such map. Let us assume that f is single-valued function defined on some domain of definition. We will treat the function as a map that maps a set from (x, y)-plane to some other set in (u, v)-plane. Let us start with two simple examples.

Example 4.12

Let $f(z) = \text{Im } z$ (defined and single-valued everywhere). Let us find its image {namely, where to the complex plane $\mathbb{C}$ is mapped}. $\text{Im } z = y \in \mathbb{R} \implies$ the complex plane $\mathbb{C}$ is mapped to $\mathbb{R}$.



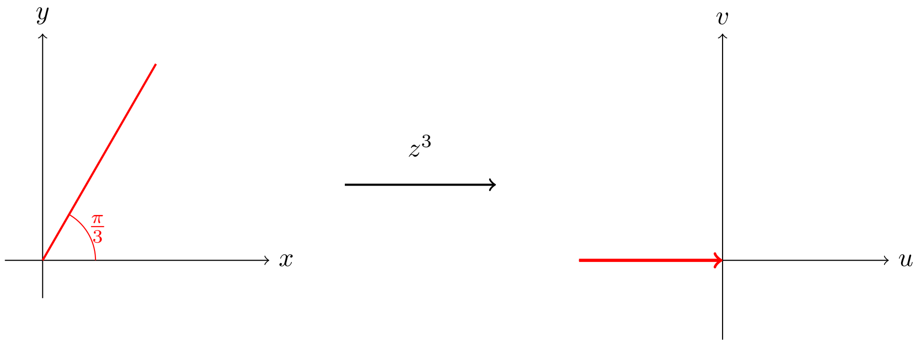
Example 4.13

What is the image of the ray $\arg z = \frac{\pi}{3}$ under $f(z) = z^3$?

Solution. Let us write the function f in polar coordinates. $f(z) = r^3(\cos\theta + i\sin\theta)^3 = r^3(\cos 3\theta + i \sin 3\theta)$ $\{r = |z|\}$ De Moivre. $\square$

Thus, the image of $\arg z = \frac{\pi}{3}$ under z^3 is $r^3(\cos(3 \cdot \frac{\pi}{3}) + i\sin(3 \cdot \frac{\pi}{3})) = -r^3 (r \geq 0!)$.

Recall that $r = |z|$, therefore $r \geq 0$ always. Hence, the image of that ray is the negative half of the real axis.

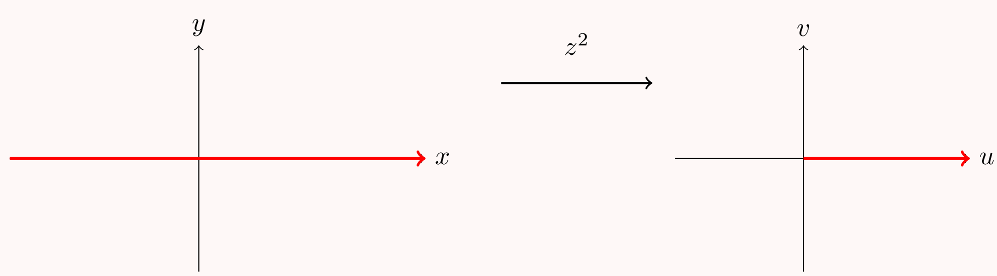


$f(z) = z^2$.

Now we will examine the mapping $f(z) = z^2$ and will find the image of various curves and shapes under this mapping. Let us rewrite f in the polar and algebraic form: $z = x + iy$: $f(x, y) = u(x, y) + iv(x, y) = x^2 - y^2 + i2xy$. $z = re^{i\theta}$: $f(r, \theta) = r^2 e^{2i\theta} = r^2 \cos 2\theta + ir^2 \sin 2\theta$.

Example 4.14

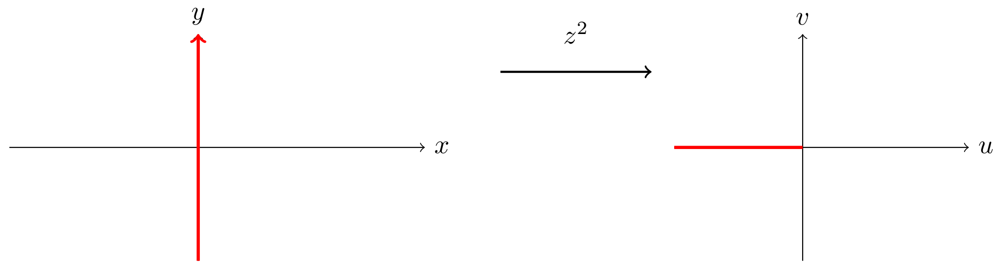
First, we determine what is the image of x-axis and of y-axis. The real axis $\mathbb{R}$ is mapped to the positive half of the real axis $\mathbb{R}^+$. $\mathbb{R} \to \mathbb{R}^+$. To see that we represent $\mathbb{R}$ as $\mathbb{R} = \mathbb{R}^+ \cup \mathbb{R}^-$, where $\mathbb{R}^+$ $(\mathbb{R}^-)$ the positive (negative) half of $\mathbb{R}$. Every point in $\mathbb{R}^+$ is of the form $re^{i0} = r \geq 0$ and every point in $\mathbb{R}^-$ is of the form $re^{i\pi} = -r \leq 0$. These points are mapped by f to: $r^2 e^{2 \cdot i \cdot 0} = (r)^2, r^2 e^{2 \cdot 0 \cdot i\pi} = r^2 \in \mathbb{R}^+$.



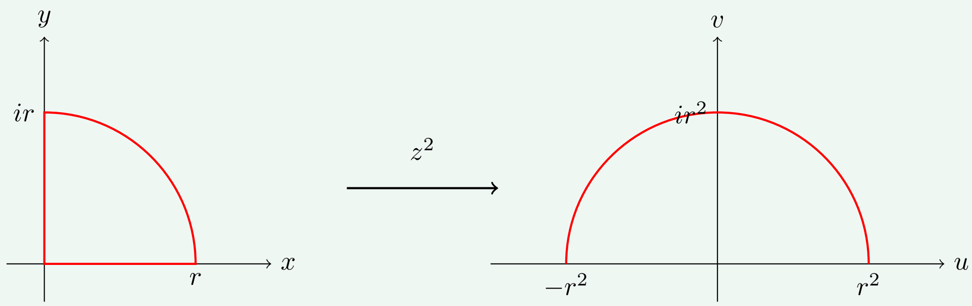
Now we will show (in the same way) that the imaginary axis is mapped to the negative half of the real axis: $i\mathbb{R} \xrightarrow{z^2} \mathbb{R}^-$.

Represent the imaginary axis $i\mathbb{R}$ as: $i\mathbb{R} = (i\mathbb{R})^+ \cup (i\mathbb{R})^-$, where $(i\mathbb{R})^+$ are points of the form $re^{i\frac{\pi}{2}} = ir (r > 0)$ and $(i\mathbb{R})^-$: $re^{i\frac{3\pi}{2}} = -ir$. f maps these points to $r^2 e^{i2 \cdot \frac{\pi}{2}} = r^2 e^{i\pi} =$

$-r^2 \leq 0, r^2e^{i2\cdot\frac{3\pi}{2}} = r^2e^{i3\pi} = -r^2 \leq 0.$

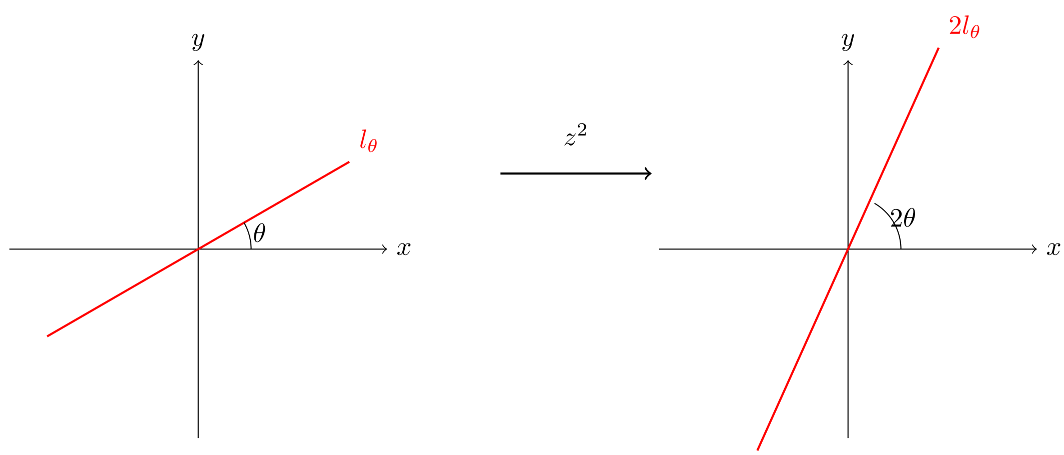


Claim 4.15 — A quarter circle of radius r centered at 0 is mapped by z^2 to a half circle of radius r^2 centered at 0.



Proof. The quarter circle is given by points $re^{i\theta}$ where $0 \leq \theta \leq \frac{\pi}{2}$. It is mapped by z^2 to $r^2e^{2i\theta}$. Since $0 \leq \theta \leq \frac{\pi}{2} \implies 0 \leq 2\theta \leq \pi$, thus we obtain a half circle of radius r^2 centered at 0. □

Claim 4.16 — A line l_θ passing through the origin at angle θ is mapped to a ray $l_{2\theta}$ starting at 0 making an angle 2θ.



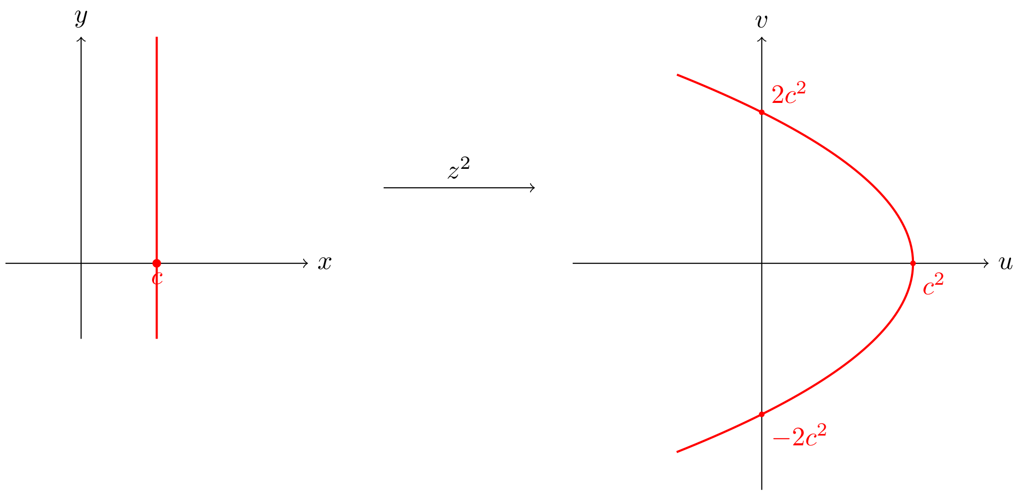
The line is parameterized by $re^{i\theta}$, where θ is fixed and r may vary. The map $f(z) = z^2$ sends r to r^2 resulting in a ray and the angle is transformed from θ to 2θ.

Now let us find the image of a vertical line $x = c \in \mathbb{R}$: In this case the geometric representation is more convenient. $z \to z^2 : (x, y) \mapsto (x^2 - y^2, 2xy)$. Set $x = c$.

We have the following system, from which we may eliminate the variable y to get an equation for a curve in (u, v)-plane:

$$u = x^2 - y^2 = c^2 - y^2$$
$$v = 2xy = 2cy$$

$y = \frac{v}{2c} \implies u = c^2 - \frac{v^2}{4c^2}$: this is leftward opening parabola with vertex $(c^2, 0)$, intersecting the v-axis at $(0, \pm 2c^2)$ {Note that: if c is negative, the image again is the same, namely, the image is independent of the sign of c}.



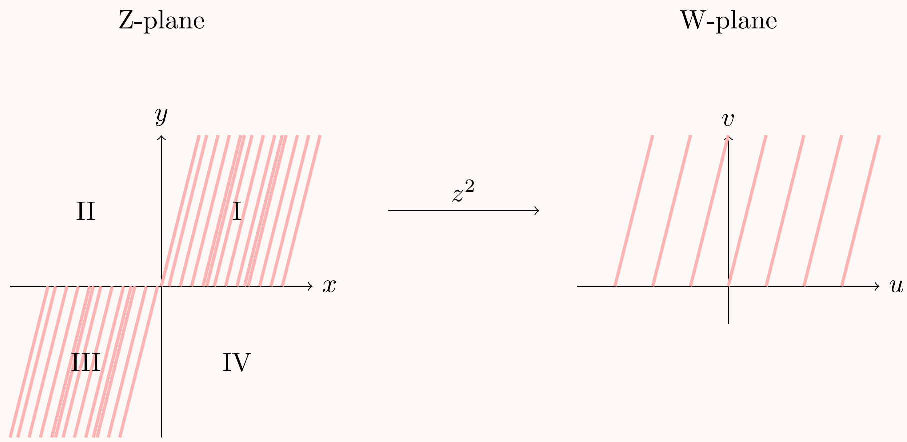
Lecture 5+6: Images under z^2

We continue to examine the images under z^2 and other transformations.

Example 4.17

Determine which regions (if any) of the z-plane are mapped to the upper half part of the w-plane under the mapping $w = z^2$.

First, we identify: the upper half of w-plane $= \{w \in \mathbb{C} | \text{Im } w \geq 0\}$ $\theta = \arg w \leq \pi$: An element $z = re^{i\theta} \to w = r^2 e^{2i\theta}$ is in the upper half plane iff the argument of w lies between 0 and π: $0 + 2\pi k \leq 2\theta \leq \pi + 2\pi k, k \in \mathbb{Z} \implies k\pi \leq \theta \leq \frac{\pi}{2} + k\pi$. Namely, z must lie in the first or third quadrant in order to map to the upper half plane [Plug $k = 1, 2, 3, 4$ to see this!]



Alternatively: we could use the algebraic form $x + iy$ and get: $w = z^2 = x^2 - y^2 + i2xy \implies \text{Im } w \geq 0 \iff 2xy \geq 0 \iff x, y \geq 0$ or $x, y \leq 0$. This gives again I and III quadrant.

Now let us study another function and its images.

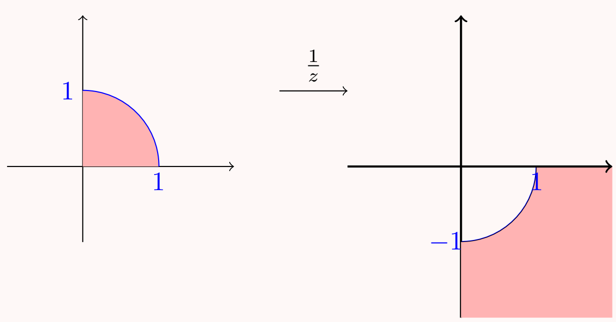
Example 4.18

Look at $z \to \frac{1}{z} = w$.

It is an instructive exercise to draw a picture trying to determine where to various regions of the plane are mapped.

EXERCISE:

1. Show that $w = \frac{1}{z}$ maps the interior of the unit circle in the first quadrant to the exterior of the unit circle in the fourth quadrant.



2. What is the image of the unit circle under $\frac{1}{z}$?

3. Write down how $z \to \frac{1}{z} = w$ maps the punctured plane $\mathbb{C} \setminus \{0\}$ to itself (and $\mathbb{C} \cup \{\infty\}$ to itself).

Example 4.19

Back to Ex 4.18. Let us study two images under $\frac{1}{z}$.

a) Find the image of the line $z = t + i(1-t)$ under $\frac{1}{z}$.

b) Determine how the horizontal lane $y = k$ is transformed under $w = \frac{1}{z}$.

Solution. a) Denote: $w = u + iv, z = x + iy \implies x = t, y = (1-t)$.

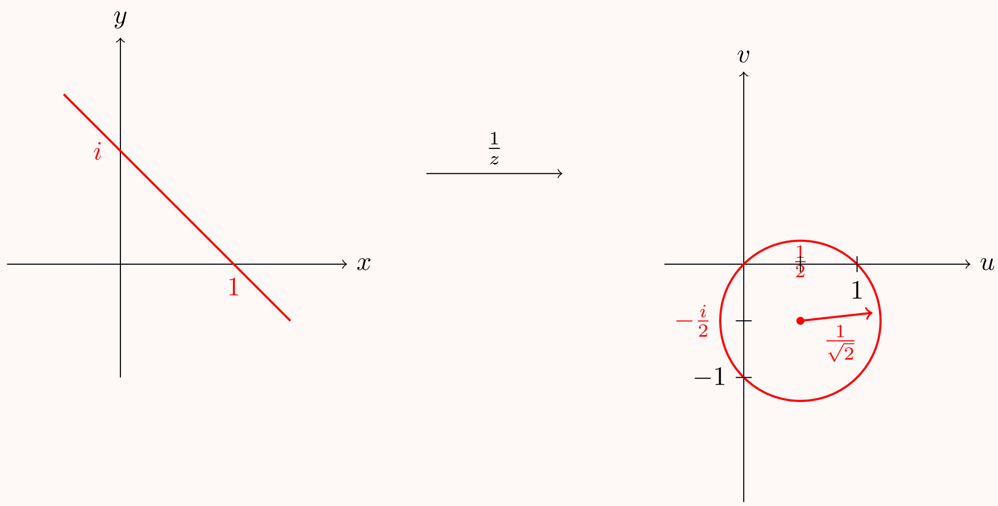
$$w = u + iv = \frac{1}{x+iy} = \frac{x-iy}{x^2+y^2}(\text{for } z \neq 0, w \neq 0) \iff u = \frac{x}{x^2+y^2}, v = \frac{-y}{x^2+y^2}$$

Thus, plugging $x = t, y = 1-t$, we obtain the following system:

$$t = \frac{u}{u^2+v^2}$$
$$1-t = \frac{-v}{u^2+v^2} \implies$$
$$t(u^2+v^2) = u$$
$$(1-t)(u^2+v^2) = -v$$

$\implies u^2+v^2 = u - v$ (we add the equations) $\implies u^2 - u + v^2 + v = 0 \implies (u-\frac{1}{2})^2 + (v+\frac{1}{2})^2 = \frac{1}{2}$ (complete the square in u and in v).

Namely, the map $w = \frac{1}{z}$ maps the line $z = t + i(1-t)$ to the circle (in w-plane) centered at the point $(\frac{1}{2}, -\frac{1}{2})$ with radius $\frac{1}{\sqrt{2}}$.

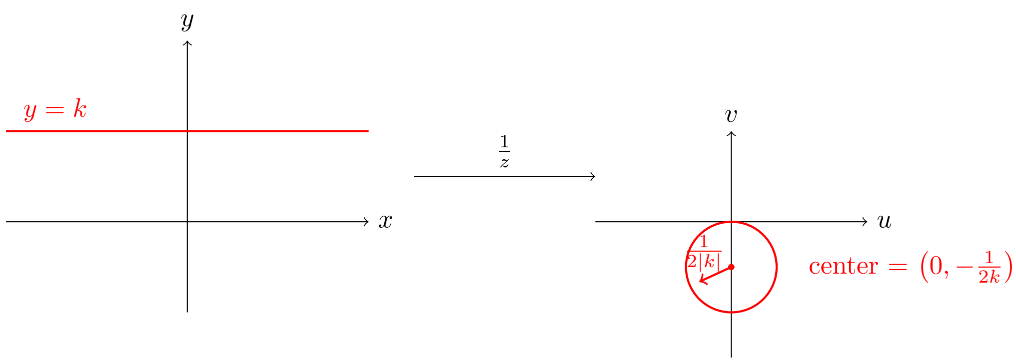


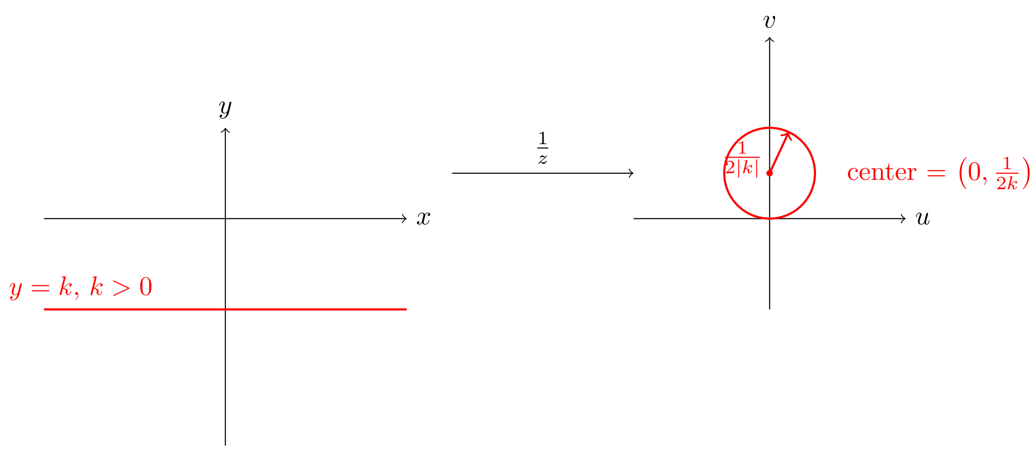
b) Substitute as before: $x = \frac{u}{u^2+v^2}, y = \frac{-v}{u^2+v^2}$. Plug $y = k$:

$$k = \frac{-v}{u^2+v^2} \iff ku^2 + kv^2 + v = 0 \iff (k \neq 0)u^2 + v^2 + \frac{v}{k} = 0$$
$$\iff u^2 + (v + \frac{1}{2k})^2 = \frac{1}{4k^2} \text{ (complete the square in } v\text{)}$$

Namely, the line is mapped onto a circle, center of which depends on the sign of k, and the radius is $\frac{1}{2|k|}$.

□





Now we have a feeling on geometrical description of the complex functions - we will study more transformations later.

Next subject - the limits of complex functions. The introduction of a limit will allow us to investigate what it means for the complex function f to be continuous and differentiable. Through several explicit examples, we contrast our findings with corresponding results for real functions.

5 Limits

5.1 Definition and properties of limits

Let f be a complex function. We say that $\lim_{z\to z_0} f(z) = w_0$ if when z is "close" to z_0, $f(z)$ is "close" to w_0. Formally:

Definition 5.1. $\lim_{z\to z_0} f(z) = w_0$ if for any given $\epsilon > 0$, there exists $\delta = \delta(\epsilon) > 0$ such that if $0 < |z - z_0| < \delta$, then $|f(z) - w_0| < \epsilon$.

Example 5.2

Let $f(z) = \frac{\bar{z}}{2}$. Claim: for any z_0, $\lim_{z\to z_0} \frac{\bar{z}}{2} = \frac{\bar{z_0}}{2}$.

Proof. Given $\epsilon > 0$, choose $\delta = 2\epsilon$. Then, if $0 < |z - z_0| < \delta$:

$$\left|\frac{\bar{z}}{2} - \frac{\bar{z_0}}{2}\right| = \left|\frac{\bar{z} - \bar{z_0}}{2}\right| = \frac{|\bar{z} - \bar{z_0}|}{2} = \frac{|\overline{z - z_0}|}{2} = \frac{|z - z_0|}{2} < \frac{\delta}{2} = \frac{2\epsilon}{2} = \epsilon$$

Thus, for a given $\epsilon > 0$ we have found δ (that depends on ϵ!) s.t. if $|z - z_0| < \delta$, then $|f(z) - w_0| = \left|\frac{\bar{z}}{2} - \frac{\bar{z_0}}{2}\right| < \epsilon$ (for any z_0!). □

Example 5.3

Let $f(z) = \frac{z}{\bar{z}}, z \neq 0$. Prove that $\lim_{z\to 0} \frac{z}{\bar{z}}$ does not exist.

Before we proceed to the proof, let us show the following:

Claim 5.4 — If $\lim_{z\to z_0} f(z)$ exists, then it is unique.

Proof. Assume $\lim_{z\to z_0} f(z) = a$ and $\lim_{z\to z_0} f(z) = b$. By Def 5.1 for any $\epsilon > 0$ there exist $\delta_1 > 0$ and $\delta_2 > 0$ such that

$$\begin{aligned} &\text{if } 0 < |z - z_0| < \delta_1 \implies |f(z) - a| < \epsilon/2 \\ &\text{if } 0 < |z - z_0| < \delta_2 \implies |f(z) - b| < \epsilon/2 \end{aligned}$$

Choose $\delta < \min(\delta_1, \delta_2) \implies$ for $0 < |z - z_0| < \delta$, $|f(z) - a| < \epsilon/2$ and $|f(z) - b| < \epsilon/2$. Thus, using triangle inequality, we obtain

$$|a - b| = |a - f(z) + f(z) - b| \le |a - f(z)| + |f(z) - b| < \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon \qquad (*)$$

Since (*) holds for any $\epsilon > 0$ we obtain that $a = b$. □

Back to Ex 5.3: we will use a common technique to show that the limit does not exist - we will show that there exist two different paths (directions) $z \to z_0$ along which we get 2 different limits.

Solution. Ex 5.2 Note: for real z we have: {namely, fix $y = 0$} $z = x \in \mathbb{R} : \frac{z}{\bar{z}} = \frac{x}{x} = 1 \implies \lim_{x\to 0} \frac{z}{\bar{z}} = 1$.

On the other hand, for purely imaginary z we get: {namely, fix $x = 0$} $z = iy \in i\mathbb{R} : \frac{z}{\bar{z}} = \frac{iy}{-iy} = -1 \implies \lim_{y\to 0} \frac{z}{\bar{z}} = -1 \implies$ the limit as $z \to 0$ does not exist. □

Note:

Remark 5.5. For any $z_0 \neq 0$, $\lim_{z\to z_0} \frac{\bar{z}}{z} = \frac{\bar{z_0}}{z_0}$.

Proof. Given $\epsilon > 0$, choose $\delta = \frac{\epsilon}{2}|z_0|$. Then

$$\begin{aligned}
&\left|\frac{\bar{z}}{z} - \frac{\bar{z_0}}{z_0}\right| = \left|\frac{\bar{z}z_0 - \bar{z_0}z}{zz_0}\right| \\
&= \left|\frac{\bar{z}z_0 + z\bar{z} - z\bar{z} - \bar{z_0}z}{zz_0}\right| = \left|\frac{\bar{z}(z_0 - z) + z(\bar{z} - \bar{z_0})}{zz_0}\right| \\
&\leq \frac{|\bar{z}||z_0 - z| + |z||\bar{z} - \bar{z_0}|}{|z||z_0|} = \frac{|z||z_0 - z|}{|z||z_0|} + \frac{|z||z - z_0|}{|z||z_0|} = \frac{2|z - z_0|}{|z_0|} < \frac{2\delta}{|z_0|} = \frac{2}{|z_0|} \cdot \frac{\epsilon}{2}|z_0| = \epsilon,
\end{aligned}$$

That's where our choice of δ come in, $\delta = \frac{\epsilon}{2}|z_0|$ □

Example 5.6

Show that $\lim_{z\to 0} \frac{\bar{z}^2}{z^2}$ does not exist.

Solution. Here, for $z = x \in \mathbb{R} : \frac{\bar{z}^2}{z^2} = \frac{x^2}{x^2} = 1$ and for $z = iy \in i\mathbb{R} : \frac{\bar{z}^2}{z^2} = \frac{(-iy)^2}{(iy)^2} = \frac{-y^2}{-y^2} = 1$. However, choosing the path $z = (1+i)t$ for $t \in \mathbb{R} \setminus \{0\}$ gives us $\frac{\bar{z}^2}{z^2} = \frac{((1-i)t)^2}{((1+i)t)^2} = \frac{(1-i)^2}{(1+i)^2} = \frac{1-2i-1}{1+2i-1} = -1$. $-1 \neq 1$ - does not agree with the value on previous two paths, thus we can conclude that the original limit does not exist. □

Alternatively: Let $z = re^{i\theta}$, $r, \theta \in \mathbb{R}$. Then $z \to 0$ means $r \to 0$. Since $e^{i\theta} \neq 0$ for all $\theta \in \mathbb{R}$, We have: $\frac{\bar{z}^2}{z^2} = \frac{r^2e^{-2i\theta}}{r^2e^{2i\theta}} = e^{-4i\theta}$. $\implies \lim_{z\to 0} \frac{\bar{z}^2}{z^2} = \lim_{r\to 0} e^{-4i\theta} = e^{-4i\theta}$. For different θ-s we get different limits - namely, on different paths the limit does not agree $\implies$ the limit does not exist.

Example 5.7

Ex 5.6 shows that the different paths need not be the real and imaginary axes!

Remark 5.8. If $\lim_{z\to z_0} f(z) = w_0$ exists, then $f(z) \to w_0$ as $z \to z_0$ on *every* path

> (uniqueness of the limit!) If on different paths $z \to z_0$ there are different values of the limit (or there exists a path on which the limit does not exist), then the limit $\lim_{z\to z_0} f(z)$ does not exist.

Similarly to the real functions we have the following proposition for the complex functions.

Proposition 5.9

If $\lim_{z\to z_0} f(z) = w_0$ and $\lim_{z\to z_0} g(z) = w_1$, then

1. $\lim_{z\to z_0}(f(z) + g(z)) = w_0 + w_1$
2. $\lim_{z\to z_0} f(z) \cdot g(z) = w_0 w_1$
3. If $w_1 \neq 0$, then $\lim_{z\to z_0} \frac{f(z)}{g(z)} = \frac{w_0}{w_1}$

Proof. EXERCISE - a very good exercise to "play" with ϵ-δ definition! □

Next proposition provides a relation between the limit of the complex function and its real and imaginary parts (that are real functions).

Proposition 5.10

Let $f(z) = u(x, y) + iv(x, y), z_0 = x_0 + iy_0$. Then $\lim_{z\to z_0} f(z) = u_0 + iv_0 = w_0 \iff \lim_{(x,y)\to(x_0,y_0)} u(x, y) = u_0 = \text{Re } w_0$ and $\lim_{(x,y)\to(x_0,y_0)} v(x, y) = v_0 = \text{Im } w_0$.

Note: the first limit is complex, but the last two limits are real!

Proof. $\implies$ Assume that $\lim_{z\to z_0} f(z) = u_0 + iv_0$. Let us show that $\lim_{(x,y)\to(x_0,y_0)} u(x, y) = u_0 = \text{Re } w_0$ and $\lim_{(x,y)\to(x_0,y_0)} v(x, y) = v_0 = \text{Im } w_0$.

By Def 5.1: given $\epsilon > 0$ there exists $\delta = \delta(\epsilon) > 0$ s.t. for all z with $0 < |z - z_0| < \delta$ we have $|f(z) - (u_0 + iv_0)| < \epsilon$.

Since for any $z \in \mathbb{C} : |\text{Im } z|, |\text{Re } z| \leq |\text{Re } z + i\text{Im } z| = |z|$, we get:

$$\begin{aligned}|v(x, y) - v_0|, |u(x, y) - u_0| < |u(x, y) - u_0 + i(v(x, y) - v_0)| &= |u(x, y) + iv(x, y) - (u_0 + iv_0)| \\ &= |f(z) - (u_0 + iv_0)| < \epsilon\end{aligned}$$

Thus $|u(x, y) - u_0| < \epsilon$ and $|v(x, y) - v_0| < \epsilon$.

Check: $|z - z_0|$ is the distance from (x, y) to (x_0, y_0) in $\mathbb{R}^2$. $\implies \lim_{(x,y)\to(x_0,y_0)} u(x, y) = u_0$ and $\lim_{(x,y)\to(x_0,y_0)} v(x, y) = v_0$.

Note: $|z - z_0| < \delta \implies |x + iy - (x_0 + iy_0)| = |(x - x_0) + i(y - y_0)| < \delta$, and we get $|x - x_0|, |y - y_0| \leq |(x - x_0) + i(y - y_0)| < \delta$.

$\impliedby$ Assume that $\lim_{(x,y)\to(x_0,y_0)} u(x, y) = u_0$ and $\lim_{(x,y)\to(x_0,y_0)} v(x, y) = v_0$.

Let $\epsilon > 0$. Since u and v have real limits u_0, v_0 as $(x, y) \to (x_0, y_0)$ from Def 5.1 there exist $\delta_1 > 0$ and $\delta_2 > 0$ such that if $0 < ||(x, y) - (x_0, y_0)|| < \delta_1$, then $|u(x, y) - u_0| < \epsilon/2$, and $0 < ||(x, y) - (x_0, y_0)|| < \delta_2$, then $|v(x, y) - v_0| < \epsilon/2$.

Choose $\delta < \min(\delta_1, \delta_2)$. Then, whenever $0 < ||(x, y) - (x_0, y_0)|| < \delta$:

$$
\begin{aligned}
|u(x, y) + iv(x, y) - (u_0 + iv_0)| &= |u(x, y) - u_0 + i(v(x, y) - v_0)| \\
&\leq |u(x, y) - u_0| + |v(x, y) - v_0| < \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon
\end{aligned}
$$

by triangle inequality.

Namely, for any $\epsilon > 0$ we have found $\delta > 0$ such that whenever $0 < |z - z_0| < \delta$: $|f(z) - (u_0 + iv_0)| < \epsilon$, as required. $\square$

Week 3

Lecture 7: Continuation from prev.

We have proved:

$$\lim_{z\to z_0} f(z) = u_0 + iv_0 \iff \lim_{(x,y)\to(x_0,y_0)} u(x,y) = u_0 \text{ and } \lim_{(x,y)\to(x_0,y_0)} v(x,y) = v_0$$

Corollary 5.11

If $\lim_{z\to z_0} f(z) = w_0$, then $\lim_{z\to z_0} |f(z)| = |w_0|$. In addition, if $w_0 \neq 0$ and $\arg w_0$ is chosen in the interval $[-\pi, \pi]$, then $\lim_{z\to z_0} \arg f(z) = \arg w_0$.

We extend the definition of limit to include the point at ∞:

5.2 Limits at infinity

Definition 5.12. We say that $\lim_{z\to\infty} f(z) = w_0$ if for any (given) $\epsilon > 0$ there exists $R = R(\epsilon) > 0$ such that for any $|z| > R$, $|f(z) - w_0| < \epsilon$.

Definition 5.13. We say that $\lim_{z\to z_0} f(z) = \infty$ if for any (given) $R > 0$ there exists $\delta = \delta(R) > 0$ such that if $0 < |z - z_0| < \delta$, then $|f(z)| > R$.

Definition 5.14. We say that $\lim_{z\to\infty} f(z) = \infty$ if for any $R > 0$ there exists $S = S(R) > 0$ such that if $|z| > S$, then $|f(z)| > R$.

Example 5.15

Prove that

a) $\lim_{z\to i} z - i = \infty$

b) $\lim_{z\to\infty} \frac{1}{z} = 0$.

Solution. a) By Def 5.13 we need to find $\delta = \delta(R) > 0$ for a given $R > 0$: If $\left|\frac{1}{z-i}\right| > R$, then $|z - i| < \frac{1}{R}$, thus the choice $\delta = \frac{1}{R}$ finishes the proof.

b) By Def 5.12 we need to find $R > 0$ for a given $\epsilon > 0$. If $\left|\frac{1}{z} - 0\right| = \frac{1}{|z|} < \epsilon$, then $|z| > \frac{1}{\epsilon}$, choose $R = \frac{1}{\epsilon}$ and finish.

□

Example 5.16

Prove that $\lim_{z\to\infty} \frac{z^2+1}{z+2i} = \infty$.

Solution. By Def 5.14 we need to find $S = S(R) > 0$ for a given $R > 0$. If $\left|\frac{z^2+1}{z+2i}\right| > R$, then

$$\left|\frac{z^2+1}{z+2i}\right| = \left|\frac{z^2(1+\frac{1}{z^2})}{z(1+\frac{2i}{z})}\right| = |z|\left|\frac{1+\frac{1}{z^2}}{1+\frac{2i}{z}}\right| > R$$

Let us show that $\left|\frac{1+\frac{1}{z^2}}{1+\frac{2i}{z}}\right| < 2$, then

$$\left|1+\frac{2i}{z}\right| \geq \left|1-\left|\frac{2i}{z}\right|\right| = \left|1-\frac{2}{|z|}\right| \geq |1-2| = 1$$

Since $z\to\infty$ we can assume that $|z| > 1$. Thus (1) $\left|1+\frac{2i}{z}\right| \geq \frac{1}{2}$.

$$\left|1+\frac{1}{z^2}\right| \leq 1 + \left|\frac{1}{z^2}\right| \leq 1+1 = 2$$

triangle inequality.

Thus (2) $\left|1+\frac{1}{z^2}\right| < 1$ and we get

$$\left|\frac{1+\frac{1}{z^2}}{1+\frac{2i}{z}}\right| = \frac{\left|1+\frac{1}{z^2}\right|}{\left|1+\frac{2i}{z}\right|} < \frac{2}{1} = 2$$

Alternatively: By Prop 5.9 2)

$$\lim_{z\to\infty}\left|\frac{z^2+1}{z+2i}\right| = \lim_{z\to\infty}|z|\left|1+\frac{1}{z^2}\right|\left|\frac{1}{1+\frac{2i}{z}}\right| = \lim_{z\to\infty}|z|\lim_{z\to\infty}\left|1+\frac{1}{z^2}\right|\lim_{z\to\infty}\left|\frac{1}{1+\frac{2i}{z}}\right|$$

By example 5.15 b): $\lim_{z\to\infty}\frac{1}{z} = 0 \implies \lim_{z\to\infty} 1+\frac{1}{z^2} = 1, \lim_{z\to\infty}\frac{1}{1+\frac{2i}{z}} = 1 \implies \lim_{z\to\infty}\frac{z^2+1}{z+2i} = \infty$. □

Our next subject is continuity of complex functions. First, we need several definitions of sets in the complex plane.

6 Sets in $\mathbb{C}$

6.1 Open sets, connectedness, and domains

Definition 6.1. A subset $S \subseteq \mathbb{C}$ is open if for each $z_0 \in S$ there exists $r > 0$ such that the disc $\{z \in \mathbb{C} | |z - z_0| < r\} \subseteq S$ is a subset of S.

Definition 6.2. An open set is connected if it is not a disjoint union of two non-empty open sets. OR: If any two points inside can be connected with a continuous line which is contained in the set.}

> **Remark 6.3.** In $\mathbb{C}$ these two definitions (in Def 6.2) are equivalent, but it is not always the case. The second - path connected - always implies the first, but there are complicated examples of sets that are connected but not path connected. OR means: For any points z, z' in the set S there exists a continuous function $p : [0, 1] \to S$ such that $p(0) = z, p(1) = z'$.

Definition 6.4. A connected open subset $R \subseteq \mathbb{C}$ is called a domain.

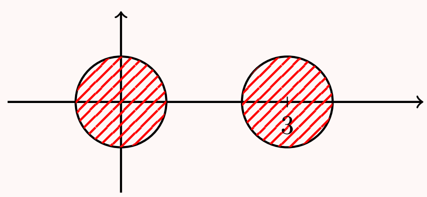
Example 6.5

a) The interior of the unit disc $\{z \in \mathbb{C} | |z| < 1\}$ is a domain - it is open and connected.

b) The unit disc with the boundary $\{z \in \mathbb{C} | |z| \leq 1\}$ is connected but not open $\implies$ it is not a domain.

Take any point on the boundary $|z| = 1$. Then, any disc around this point is not contained in the unit disc regardless how small its radius.

c) $\{z \in \mathbb{C} | |z| < 1\} \cup \{z \in \mathbb{C} | |z - 3| < 1\}$ - open, but not connected -



- it is a union of 2 non-empty disjoint open sets.

6.2 Interior, boundary, and closed sets

Definition 6.6. Let $S \subseteq \mathbb{C}$. $z_0 \in S$ is an interior point of S if there exists $r > 0$ such that the disc $\{z \in \mathbb{C} | |z - z_0| < r\} \subseteq S$. $z_1 \in S$ is a boundary point of S if every open disc around it, $\{z \in \mathbb{C} | |z - z_1| < r\}$ for any $r > 0$ contains points from S and not from S ($\mathbb{C} \setminus S$). The collection of boundary points of S is called the boundary of S and denoted by ∂S.

An equivalent definition of an open set:

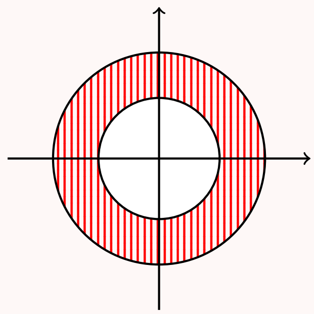
Definition 6.7. $\frac{1}{2}$ $S \subseteq \mathbb{C}$ is open if it contains only interior points.

Definition 6.8. A set $S \subseteq \mathbb{C}$ is closed if it contains all its boundary points.

Let $S \subseteq \mathbb{C}$ be an open set. If we add to S all its boundary points we always will get a closed set! The closure of S: $\bar{S} = S \cup \partial S$ is a closed set.

Remark 6.9. There exist sets that are neither open nor closed!

Example 6.10 1. The set $S = \{z \in \mathbb{C} | 1 < |z| < 4\}$ is an open set. Its boundary consists of two parts:



$\partial S = \{z \in \mathbb{C} || z| = 1, |z| = 4\}$ and its closure is: $\bar{S} = S \cup \partial S = \{z \in \mathbb{C} | 1 \leq |z| \leq 4\}$.

2. The set $S = \{z \in \mathbb{C} | 1 \leq |z| < 4\}$ neither open nor closed : it contains only part of its boundary points $\{z \in \mathbb{C} || z| = 1\}$ but not all the boundary points, as $\{z \in \mathbb{C} || z| = 4\}$ is not in S.

6.3 Neighborhoods

Definition 6.11. A collection of all the points $z \in \mathbb{C}$ that are on the distance $\epsilon > 0$ from z_0 is called ϵ-neighborhood of $z_0 \in \mathbb{C}$. For $\epsilon > 0, z_0 \in \mathbb{C}$: ϵ-neighborhood of z_0 is the set $\{z \in \mathbb{C} || z - z_0| < \epsilon\}$.

Having all these definitions, now we can study the continuity of a complex function.

7 Continuity

7.1 Definition and basic properties

Definition 7.1. The function $f(z)$ is continuous at the point z_0 if it is defined at a neighborhood of z_0 (and at z_0) and $\lim_{z \to z_0} f(z) = f(z_0)$. Namely: f is defined at z_0, at some ϵ-neighborhood of z_0, the limit above exists and it is equal to the value of the function at the point z_0. The function $f(z)$ is continuous in some domain S if it is continuous at every point $z \in S$.

As a consequence of the limit laws in Prop 3.1, we have the following (similar) proposition about the continuity.

Proposition 7.2

If $f(z)$ and $g(z)$ are continuous at $z_0 \in \mathbb{C}$, then so are $f+g$ and $f \cdot g$, and if $g(z_0) \neq 0$, then $\frac{f}{g}$ is continuous at z_0.

We also have an analogue of Prop 3.2:

Proposition 7.3

The function $f(z) = u(x,y) + iv(x,y)$ is continuous at the point $z_0 = x_0 + iy_0 \iff$ the real functions $u(x,y), v(x,y)$ are continuous at the point (x_0, y_0).

Let us see some examples of continuous functions.

Example 7.4

a) It is obvious that the constant function is continuous everywhere. $f(z) = c$ is continuous at every $z \in \mathbb{C}$.

b) It is also clear that the identity function is continuous everywhere: $f(z) = z$ is continuous at every $z \in \mathbb{C}$ (**Check!**- from Def 7.1!)

c) Combining a), b), and Prop 5.1: We conclude that the linear function is continuous everywhere: for constants $a, b \in \mathbb{C}$, $f(z) = az + b$ is continuous at every $z \in \mathbb{C}$.

Example 7.5

Prove that $z^n, n \in \mathbb{N}$ is continuous for all $z \in \mathbb{C}$.

Proof. We prove it by checking the definition Def 5.1 at some $z_0 \in \mathbb{C}$. Consider:

$$z^n - z_0^n = (z - z_0)(z^{n-1} + z^{n-2}z_0 + z^{n-3}z_0^2 + \cdots + zz_0^{n-2} + z_0^{n-1})$$

Denote $|z| = r, |z_0| = r_0$. Then, we obtain

$$|z^n - z_0^n| \leq |z - z_0|(r^{n-1} + r^{n-2}r_0 + \cdots + rr_0^{n-2} + r_0^{n-1})$$

Let us look at all the z-s inside the disc of radius δ centered at $z_0 : \{z \in \mathbb{C} | |z - z_0| < \delta\}$. For every z in this set $r = |z| = |z - z_0 + z_0| \leq |z - z_0| + |z_0| < \delta + r_0$, thus

$$\begin{aligned} |z^n - z_0^n| &\leq |z - z_0|((r_0 + \delta)^{n-1} + (r_0 + \delta)^{n-2}r_0 + \cdots + (r_0 + \delta)r_0^{n-2} + r_0^{n-1}) \\ &\leq |z - z_0| n (r_0 + \delta)^{n-1} \end{aligned}$$

The choice of δ small enough so that $|z^n - z_0^n|$ is smaller than the given ϵ finishes the proof. □

Combining Prop (7.2), Ex (7.5), and Ex (7.4) a),b), we conclude:

Proposition 7.6

5.3 Every polynomial $P_n(z) = a_0 + a_1 z + a_2 z^2 + \cdots + a_n z^n$ with complex coefficients $a_0, a_1, a_2, a_3, \ldots, a_n \in \mathbb{C}$ is continuous at every $z \in \mathbb{C}$. Furthermore, every rational function $f(z) = \frac{P_n(z)}{Q_m(z)}$ (P_n, Q_m polynomials of degree n, m respectively) is continuous at every $z \in \mathbb{C}$, except possibly at the roots of $Q_m(z)$.

"Possibly" since if there is a common root to $P_n(z)$ and $Q_m(z)$ that can be factorized, the function *may* be continuous at that root. We said that rational functions are continuous except possibly at the roots of the denominator, however, recalling Def 5.13: Def 5.13 $\implies$ Rational functions can be regarded as continuous everywhere on $\mathbb{C} \cup \{\infty\}$.

In practice, one of the best ways of dealing with "∞" in limits is to use substitution $z = \frac{1}{\zeta}$ and let $\zeta \to 0$.

Example 7.7

1. $\lim_{z\to -1} \frac{iz+3}{z+1} = \infty$ since $\lim_{z\to -1} \frac{z+1}{iz+3} = \frac{-1+1}{-i+3} = 0$.
2. $\lim_{z\to\infty} \frac{z+2i}{z^2+8i} = \lim_{\zeta\to 0} \frac{\frac{1}{\zeta}+2i}{\frac{1}{\zeta^2}+8i} = \lim_{\zeta\to 0} \frac{\zeta(1+2i\zeta)}{1+8i\zeta^2} = \lim_{\zeta\to 0} \zeta \lim_{\zeta\to 0} \frac{1+2i\zeta}{1+8i\zeta^2} = 0 \cdot 1 = 0$.
3. $\lim_{z\to\infty} \frac{5z+i}{z+i} = \lim_{\zeta\to 0} \frac{\frac{5}{\zeta}+i}{\frac{1}{\zeta}+i} = \lim_{\zeta\to 0} \frac{5+i\zeta}{1+i\zeta} = \lim_{\zeta\to 0}(5+i\zeta) \lim_{\zeta\to 0} \frac{1}{1+i\zeta} = 5 \cdot 1 = 5$.

Example 7.8

Is $\frac{z^3+8i}{z+2i}$ continuous at $z = 2i$?

Solution. Since $2i + 1 \neq 0$, by Prop 7.2:

$$\lim_{z\to 2i} \frac{z^3+8i}{z+2i} = \frac{\lim_{z\to 2i}(z^3+8i)}{\lim_{z\to 2i} z+1} = \frac{(8i^3+8i)}{2i+1} = \frac{-8i+8i}{2i+1} = 0,$$

which $\implies$ continuous. The function is defined at $z = 2i$ and in a neighborhood of $z = 2i$, the limit exists and equal to the value of the function $\implies$ the function is continuous at $z = 2i$. □

Lecture 8+9: Introduction to Differentiability in Complex...

We have studied the limits and the continuity of complex functions. Now we will study what does it mean for the complex function to be differentiable.

8 Differentiability

8.1 Definition and examples

Definition 8.1. Let $f(z)$ be defined in a neighborhood of a point $z_0 \in \mathbb{C}$. If the limit $\lim_{z\to z_0} \frac{f(z)-f(z_0)}{z-z_0}$ exists, then f is differentiable at z_0 and $f'(z_0) = \lim_{z\to z_0} \frac{f(z)-f(z_0)}{z-z_0}$.

Alternative notation: Denote $\Delta z = z - z_0$. Then $f'(z_0) = \lim_{\Delta z\to 0} \frac{f(z_0+\Delta z)-f(z_0)}{\Delta z}$.

Example 8.2

Using Def 6.1 compute the derivative of

a) $f(z) = z$

b) $f(z) = z^2$

Solution. a)

$$\lim_{\Delta z\to 0} \frac{f(z_0+\Delta z)-f(z_0)}{\Delta z} = \lim_{\Delta z\to 0} \frac{z_0+\Delta z - z_0}{\Delta z} = \lim_{\Delta z\to 0} \frac{\Delta z}{\Delta z} = 1,$$

which $\implies$ $f'(z)$ exists for all z and $f'(z) = 1$.

b)

$$\begin{aligned}
&\lim_{\Delta z\to 0} \frac{f(z_0+\Delta z)-f(z_0)}{\Delta z} = \lim_{\Delta z\to 0} \frac{(z_0+\Delta z)^2 - z_0^2}{\Delta z} \\
&= \lim_{\Delta z\to 0} \frac{z_0^2 + 2z_0\Delta z + \Delta z^2 - z_0^2}{\Delta z} = \lim_{\Delta z\to 0} \frac{\Delta z(2z_0+\Delta z)}{\Delta z} = 2z_0 + \lim_{\Delta z\to 0} \Delta z = 2z_0,
\end{aligned}$$

which $\implies$ $f'(z)$ exists for all z and $f'(z) = 2z$.

□

Example 8.3

8.2 Prove that for any $n \in \mathbb{N}$, $(z^n)' = nz^{n-1}$ for all $z \in \mathbb{C}$.

Proof. Recall: $(a+b)^n = \sum_{k=0}^n \binom{n}{k} a^k b^{n-k}$, where $\binom{n}{k} = \frac{n!}{k!(n-k)!}$.

$$\begin{aligned}
&\lim_{\Delta z\to 0} \frac{f(z_0+\Delta z)-f(z_0)}{\Delta z} = \lim_{\Delta z\to 0} \frac{(z_0+\Delta z)^n - z_0^n}{\Delta z} \\
&= \lim_{\Delta z\to 0} \frac{z_0^n + nz_0^{n-1}\Delta z + \frac{n(n-1)}{2} z_0^{n-2}\Delta z^2 + \cdots + \Delta z^n - z_0^n}{\Delta z} \\
&= \lim_{\Delta z\to 0} \left(nz_0^{n-1} + \frac{n(n-1)}{2} z_0^{n-2}\Delta z + \cdots + \Delta z^{n-1} \right) = nz_0^{n-1}
\end{aligned}$$

$\implies f'(z)$ exists for all z and $f'(z) = nz^{n-1}$. □

These examples suggest that the differentiation of complex functions is similar to the differentiation of real functions. Let us see examples that it may be very different!

Example 8.4
Compute the derivative of $f(z) = \bar{z}$.

Solution.

$$\lim_{\Delta z \to 0} \frac{f(z_0 + \Delta z) - f(z_0)}{\Delta z} = \lim_{\Delta z \to 0} \frac{\overline{z_0 + \Delta z} - \bar{z}_0}{\Delta z} = \lim_{\Delta z \to 0} \frac{\bar{z}_0 + \overline{\Delta z} - \bar{z}_0}{\Delta z} = \lim_{\Delta z \to 0} \frac{\overline{\Delta z}}{\Delta z}$$

- doesn't exist! (Ex 5.2) □

We have proved (Ex 5.2) that this limit does not exist by looking at the limits along real and imaginary axes and showing that they do not agree. $\implies f(z) = \bar{z}$ is not differentiable at any $z \in \mathbb{C}$.

Example 8.5
8.4 Compute the derivative of $f(z) = |z|^2$.

Solution.

$$\begin{aligned}
\lim_{\Delta z \to 0} \frac{|z + \Delta z|^2 - |z|^2}{\Delta z} &= \lim_{\Delta z \to 0} \frac{(z + \Delta z)(\bar{z} + \overline{\Delta z}) - |z|^2}{\Delta z} \\
&= \lim_{\Delta z \to 0} \frac{z\bar{z} + \Delta z \bar{z} + z\overline{\Delta z} + \Delta z \overline{\Delta z} - z\bar{z}}{\Delta z} \\
&= \lim_{\Delta z \to 0} \left(\bar{z} + z\frac{\overline{\Delta z}}{\Delta z} + \overline{\Delta z} \right) \qquad (*)
\end{aligned}$$

If $z \neq 0$, then $\lim_{\Delta z \to 0} \frac{\overline{\Delta z}}{\Delta z}$ does not exist! (tends to z along the real axis and to $(-z)$ along imaginary axis $\implies$ does not exist) $\implies$ The limit (*) does not exist $\implies f'(z)$ does not exist.

If $z = 0 : \lim_{\Delta z \to 0} \frac{\overline{\Delta z}}{\Delta z} \cdot 0 = 0, \bar{z} = 0 \implies$ the limit (*) exists and $f'(0) = 0$.

Thus, $f(z) = |z|^2$ is differentiable only at $z = 0$. □

Now we observe how we can construct complicated differentiable functions from elementary ones. The following proposition is analogous to the real case.

8.2 Differentiation rules

Proposition 8.6

If $c \in \mathbb{C}$ is a constant and $f(z), g(z)$ are differentiable at z, then:

1. $c' = 0$
2. $(cf(z))' = cf'(z)$
3. $f + g$ is differentiable at z and $(f+g)'(z) = f'(z) + g'(z)$
4. $f \cdot g$ is differentiable at z and $(fg)'(z) = f'(z)g(z) + f(z)g'(z)$
5. If $g(z) \neq 0$, then $\frac{f}{g}$ is differentiable at z and $\left(\frac{f}{g}\right)'(z) = \frac{g(z)f'(z) - f(z)g'(z)}{(g(z))^2}$
6. Chain rule. If $f(z)$ is differentiable at z_0 and $g(z)$ is differentiable at $w_0 = f(z_0)$, then the composed function $(g \circ f)(z) = g(f(z))$ is differentiable at z_0 and $(g \circ f)'(z_0) = g'(w_0)f'(z_0) = g'(f(z_0))f'(z_0)$.

Exercise - on your own.

Exercise 8.7. (The details follow from Def 8.1, Prop 5.9; the chain rule - proof follows from Def 8.1 as in the real case).

□

We have seen that the constant function, the identity function and z^n are differentiable everywhere. Since any polynomial can be constructed out of these functions, using Prop 6.1 we conclude:

Combining Ex 8.1, Ex 8.2, and Prop 6.1 1), 3), we obtain:

Corollary 8.8

6.1. Any polynomial $P_n(z) = a_0 + a_1 z + \cdots + a_n z^n, a_0, \ldots, a_n \in \mathbb{C}$ is differentiable everywhere in $\mathbb{C}$ and $(P_n)'(z) = a_1 + 2a_2 z + 3a_3 z^2 + \cdots + na_n z^{n-1}$.

Prop 6.1 with Cor 6.1 imply

Corollary 8.9

6.2 Any rational function $f(z) = \frac{P_n(z)}{Q_m(z)}$ (P_n, Q_m polynomials of degree n, m respectively) is differentiable everywhere in $\mathbb{C}$, except possibly at the roots of $Q_m(z)$, and $f'(z) = \frac{P_n'(z)Q_m(z) - P_n(z)Q_m'(z)}{(Q_m(z))^2}, (Q_m(z) \neq 0)$.

Example 8.10

8.5 Compute the derivative of $f(z) = \frac{z^3-5z+2i}{z^2-3iz}$.

Solution. Combining Prop 6.4, Cor 6.2, and Ex 8.2 we obtain

$$f'(z) = \frac{(3z^2-5)(z^2-3iz)-(z^3-5z+2i)(2z-3i)}{(z^2-3iz)^2} = \frac{z^4-6iz^3+5z^2-4iz-6}{(z^2-3iz)^2}$$

(exists for $z \neq 0, 3i$, for which $z^2 - 3iz = 0$ and the numerator is not 0!) □

Next proposition provides a relation between the continuity and differentiability.

Proposition 8.11

If f is differentiable at z_0, then f is continuous at z_0.

Proof. Observe that

$$\begin{aligned}\lim_{z\to z_0}(f(z)-f(z_0)) &= \lim_{z\to z_0}\frac{f(z)-f(z_0)}{z-z_0}(z-z_0)\\ &= \lim_{z\to z_0}\frac{f(z)-f(z_0)}{z-z_0}\lim_{z\to z_0}(z-z_0) = f'(z_0)\cdot 0 = 0\\ &\implies \lim_{z\to z_0}(f(z)-f(z_0)) = (\lim_{z\to z_0} f(z)) - f(z_0) = 0 \iff \lim_{z\to z_0} f(z) = f(z_0)\end{aligned}$$

□

Remark 8.12. The converse is false! $f(z) = \bar{z}$ is continuous everywhere, yet differentiable nowhere!

This leads to the following natural question: Q-n: What are the necessary and sufficient conditions for a complex function to be differentiable?

8.3 Cauchy-Riemann equations

We are going to state and partially prove these conditions. Recall: A real function $u(x, y)$ is differentiable at the point (x_0, y_0) if both partial derivatives $\frac{\partial u}{\partial x}, \frac{\partial u}{\partial y}$ exist at (x_0, y_0), and

$$\lim_{(x,y)\to(x_0,y_0)} \frac{u(x,y) - u(x_0,y_0) - \frac{\partial u}{\partial x}(x_0,y_0)(x-x_0) - \frac{\partial u}{\partial y}(x_0,y_0)(y-y_0)}{\sqrt{(x-x_0)^2+(y-y_0)^2}} = 0$$

Namely, the function $\left(u(x_0,y_0) + \frac{\partial u}{\partial x}(x_0,y_0)(x-x_0) + \frac{\partial u}{\partial y}(x_0,y_0)(y-y_0)\right)$ is a good approximation to $u(x, y)$ near the point (x_0, y_0). In particular, it means that: There exists a neighborhood of (x_0, y_0) s.t. $\frac{\partial u}{\partial x}, \frac{\partial u}{\partial y}$ are defined everywhere in this neighborhood and continuous at (x_0, y_0).

Now we are ready to state the conditions.

Theorem 8.13

The function $f(z) = u(x, y) + iv(x, y)$ is differentiable at $z_0 = x_0 + iy_0$ if and only if $u(x, y)$ and $v(x, y)$ are differentiable at (x_0, y_0) and the following conditions, called Cauchy-Riemann equations, hold:

$$\begin{aligned} \frac{\partial u}{\partial x}(x_0, y_0) &= \frac{\partial v}{\partial y}(x_0, y_0) \\ \frac{\partial u}{\partial y}(x_0, y_0) &= -\frac{\partial v}{\partial x}(x_0, y_0) \quad \text{CR equations} \end{aligned}$$

Furthermore: $f'(z_0) = \frac{\partial u}{\partial x}(x_0, y_0) + i\frac{\partial v}{\partial x}(x_0, y_0)$.

The contrapositive statement: if Cauchy-Riemann equations fail, then the function is not differentiable. However, it is **not** true that if CR equations hold, then the function is differentiable, namely, we cannot drop the assumption that u and v are differentiable at (x_0, y_0).

Example 8.14

8.6 Let $z = x + iy, f(z) = \sqrt{|xy|}$, namely: $u(x, y) = \sqrt{|xy|}, v(x, y) = 0$.

Check: $f(z)$ satisfies CR equations at $z = 0$:

$$\frac{\partial u}{\partial x}(0, 0) = \frac{\partial v}{\partial y}(0, 0) = \frac{\partial u}{\partial y}(0, 0) = \frac{\partial v}{\partial x}(0, 0) = 0$$

Check: $u(x, y)$ is not differentiable at $(0, 0)$.

Claim 8.15 — $f'(0)$ does not exist.

Proof. Need to show that $\lim_{z\to 0} \frac{f(z)-f(0)}{z-0} = \lim_{z\to 0} \frac{f(z)}{z}$ does not exist.

We do that by showing that on different paths the limit takes different values.

Let $x = \alpha r, y = \beta r$, where $\alpha, \beta \in \mathbb{R}, \alpha, \beta \neq 0$ constants. Assume $r > 0$ ($r < 0$ in the same way).

$$\lim_{z\to 0} \frac{f(z)}{z} = \lim_{(x,y)\to(0,0)} \frac{\sqrt{|xy|}}{x + iy} = \lim_{r\to 0} \frac{\sqrt{|\alpha\beta| r^2}}{r(\alpha + i\beta)} = \lim_{r\to 0} \frac{\sqrt{|\alpha\beta|} r}{r(\alpha + i\beta)} = \frac{\sqrt{|\alpha\beta|}}{\alpha + i\beta}$$

Namely, for different α and β the limit takes different values, thus does not exist. $\square$

This example shows that it is not enough for the sufficient part that the partial derivatives are just defined at a point, they need to be defined in a neighborhood and to be continuous at the point. For the necessary part, it is enough though, and we will prove this direction now.

First, we state a corollary that gives polar form of Cauchy-Riemann equations.

Corollary 8.16 (Polar form of Cauchy-Riemann equations)

Let $z = re^{i\theta} = r(\cos\theta + i\sin\theta)$, where $r = |z|, \theta = \arg z \mod 2\pi$. Let $f(z) = \psi(r,\theta) + i\phi(r,\theta)$. Then $\frac{\partial\psi}{\partial r} = \frac{1}{r}\frac{\partial\phi}{\partial\theta}$ and $\frac{\partial\phi}{\partial r} = -\frac{1}{r}\frac{\partial\psi}{\partial\theta}$. Moreover: $f'(z) = e^{-i\theta}(\frac{\partial\psi}{\partial r} + i\frac{\partial\phi}{\partial r})$.

Now we prove the following form of the necessary condition:

Proposition 8.17

If $f(z) = u(x,y) + iv(x,y)$ is differentiable at $z_0 = x_0 + iy_0$, then $\frac{\partial u}{\partial x}, \frac{\partial u}{\partial y}, \frac{\partial v}{\partial x}, \frac{\partial v}{\partial y}$ all exist at (x_0,y_0), and $\frac{\partial u}{\partial x}(x_0,y_0) = \frac{\partial v}{\partial y}(x_0,y_0)$ and $\frac{\partial u}{\partial y}(x_0,y_0) = -\frac{\partial v}{\partial x}(x_0,y_0)$.

This statement is weaker as we are not proving that u and v are differentiable at (x_0,y_0).

Proof. Since f is differentiable at z_0, the following limit exists:

$$\begin{aligned}
f'(z_0) &= \lim_{\Delta z\to 0}\frac{f(z_0+\Delta z) - f(z_0)}{\Delta z}\\
&= \lim_{\substack{\Delta x\to 0\\ \Delta y\to 0}}\frac{u(x_0+\Delta x, y_0+\Delta y) + iv(x_0+\Delta x, y_0+\Delta y) - (u(x_0,y_0) + iv(x_0,y_0))}{\Delta x + i\Delta y}\\
&= \lim_{\substack{\Delta x\to 0\\ \Delta y\to 0}}\left[\frac{u(x_0+\Delta x, y_0+\Delta y) - u(x_0,y_0) + i(v(x_0+\Delta x, y_0+\Delta y) - v(x_0,y_0))}{\Delta x + i\Delta y}\right] \qquad (*)
\end{aligned}$$

Since the limit (*) exists, it is unique and independent of path $\Delta z \to 0$. We compute this limit along 2 different paths:

1. $\Delta x \to 0, \quad \Delta y = 0$
2. $\Delta x = 0, \quad \Delta y \to 0$

1.

$$\begin{aligned}
f'(z_0) &= \lim_{\substack{\Delta x\to 0\\ \Delta y= 0}}\left[\frac{u(x_0+\Delta x, y_0) - u(x_0,y_0) + i(v(x_0+\Delta x, y_0) - v(x_0,y_0))}{\Delta x}\right]\\
&= \frac{\partial u}{\partial x}(x_0,y_0) + i\frac{\partial v}{\partial x}(x_0,y_0)
\end{aligned}$$

2.

$$f'(z_0) = \lim_{\substack{\Delta x = 0 \\ \Delta y \to 0}} \left[\frac{u(x_0, y_0 + \Delta y) - u(x_0, y_0) + i(v(x_0, y_0 + \Delta y) - v(x_0, y_0))}{i\Delta y} \right]$$
$$= \frac{1}{i}\frac{\partial u}{\partial y}(x_0, y_0) + \frac{\partial v}{\partial y}(x_0, y_0) = -i\frac{\partial u}{\partial y}(x_0, y_0) + \frac{\partial v}{\partial y}(x_0, y_0)$$

Since on both paths we get the same limit, we conclude that

$$\frac{\partial u}{\partial x}(x_0, y_0) + i\frac{\partial v}{\partial x}(x_0, y_0) = -i\frac{\partial u}{\partial y}(x_0, y_0) + \frac{\partial v}{\partial y}(x_0, y_0).$$

Comparing the real and imaginary parts we obtain

$$\frac{\partial u}{\partial x}(x_0, y_0) = \frac{\partial v}{\partial y}(x_0, y_0) \text{ and } -\frac{\partial u}{\partial y}(x_0, y_0) = \frac{\partial v}{\partial x}(x_0, y_0).$$

□

Example 8.18

Where

$$f(x, y) = x^3 - 3xy^2 + i(3x^2y - y^3)$$

is differentiable?

Solution. Let us check where $u = x^3 - 3xy^2, v = 3x^2y - y^3$ are differentiable and CR equations hold.

$$\frac{\partial u}{\partial x} = 3x^2 - 3y^2 \qquad \frac{\partial v}{\partial x} = 6xy$$
$$\frac{\partial u}{\partial y} = -6xy \qquad \frac{\partial v}{\partial y} = 3x^2 - 3y^2$$

$\Longrightarrow$ u, v are differentiable at every (x, y). CR equations:

$$\begin{cases} \frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \\ \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x} \end{cases} \iff \begin{cases} 3x^2 - 3y^2 = 3x^2 - 3y^2 \to \text{holds only for } x = 0 \\ -6xy = -6xy \to \text{since } x = 0, \text{holds for any } y \end{cases}$$

$\Longrightarrow$ $f(z)$ is differentiable at all points of the form $(0, y)$ and $f'(0, y) = \frac{\partial u}{\partial x}(0, y) + i\frac{\partial v}{\partial x}(0, y) = -3y^2$. □

Week 4

Lecture 10: Analyticity and other...

We have studied the continuity and differentiability of complex functions. Our next subject - analytic functions.

9 Single-valued analytic functions.

Recall: If for any z corresponds one value $f(z)$, then we say that f is single-valued function.

9.1 Definitions and examples

Definition 9.1. Single-valued function $f(z)$ is analytic at the point z_0 if there exists ϵ-neighborhood $D(z_0, \epsilon)$ of z_0 so that f is differentiable at every $z \in D(z_0, \epsilon)$.

We have proved that if f is differentiable at z_0, then f is continuous at z_0, therefore if f is analytic at z_0, then f is continuous at z_0.

Let us note that we will also use names Holomorphic or Regular functions for analytic functions.

Definition 9.2. A function that is analytic for every $z \in \mathbb{C}$ (on the whole plane) is called an entire function.

Definition 9.3. Single-valued function is analytic on a domain (open connected subset of the plane $\mathbb{C}$) Ω if it is differentiable at every $z \in \Omega$.

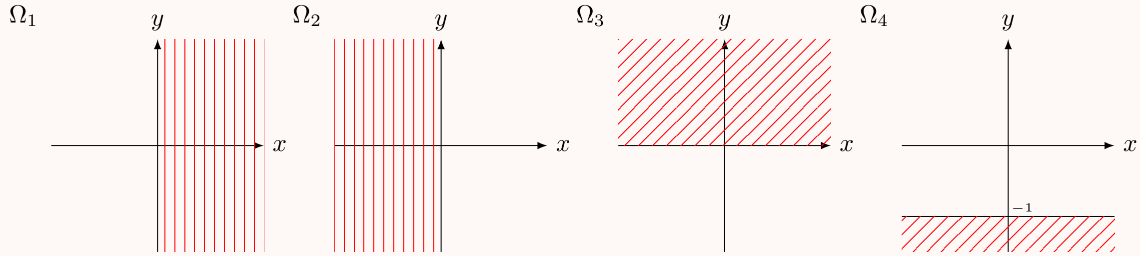
Note: From Def 9.1: a set on which the function is analytic needs to be open, however if we say that some function f is analytic in a closed set, we mean that there exists an open set that contains this closed set and the function is analytic on this open set.

Example 9.4

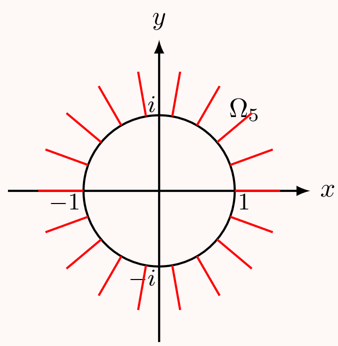
Polynomial $P_n(z) = a_0 + a_1 z + \cdots + a_n z^n$ - single-valued function differentiable at every $z \in \mathbb{C} \implies$ analytic on $\mathbb{C} \implies$ entire.

Example 9.5

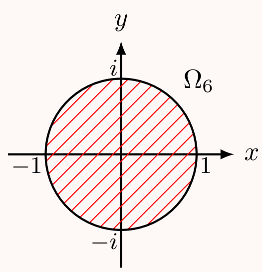
Every rational function $f(z) = \frac{P_n(z)}{Q_m(z)}$ where $P_n(z), Q_m(z)$ are polynomials (of deg n, m respectively) without common factors is analytic on $\mathbb{C}$ except for the roots of $Q_m(z)$, namely except for the points where $Q_m(z) = 0$. For instance: $f(z) = \frac{z-1}{z^2+iz} = \frac{z-1}{z(z+i)}$ is analytic everywhere except for $z = 0, -i$. For example, it is analytic on the following domains:



$\{z \in \mathbb{C} : \mathrm{Re}\, z > 0\}$ $\{z \in \mathbb{C} : \mathrm{Re}\, z < 0\}$ $\{z \in \mathbb{C} : \mathrm{Im}\, z > 0\}$ $\{z \in \mathbb{C} : \mathrm{Im}\, z < -1\}$



$\{z \in \mathbb{C} : |z| > 1\}$



$\{z \in \mathbb{C} : |z| < 1\}$

$\{z \in \mathbb{C} : \mathrm{Re}\ z > 0\}$ $\{z \in \mathbb{C} : \mathrm{Re}\ z < 0\}$ $\{z \in \mathbb{C} : \mathrm{Im}\ z > 0\}$ $\{z \in \mathbb{C} : \mathrm{Im}\ z < -1\}$

But it is not analytic in Ω_6, since $\partial \in \{z \in \mathbb{C} : |z| < 1\}$ and f is not defined at $z = 0$. $\{z \in \mathbb{C} : |z| > 1\}$ $\{z \in \mathbb{C} : |z| < 1\}$

Note: f is analytic on a domain $\Omega \iff u$ and v are differentiable on Ω and CR equations hold.

Exercise (Ex 9.3) The function $f(z) = e^x \cos y + ie^x \sin y = e^z$ is entire.

In the next example: a function for which CR equations hold on the whole plane, but u and v are *not* differentiable on the whole plane, thus the function is ***not*** entire.

Example 9.6

$$f(z) = \begin{cases} e^{-1/z^4} & z \neq 0 \\ 0 & z = 0 \end{cases} \text{ defined everywhere in } \mathbb{C}$$

Check: CR equations hold for every $z \in \mathbb{C}$.

Claim:

Claim 9.7 — $f(z)$ is not continuous at $z = 0$ ($\implies$ not differentiable).

Proof. Compute $\lim_{z\to 0} f(z)$ on the path $y = x$:

$$\lim_{\substack{x\to 0\\ y=x}} e^{-\frac{1}{(x+iy)^4}} = \lim_{x\to 0} e^{-\frac{1}{(1+i)^4 x^4}} = \lim_{x\to 0} e^{\frac{1}{4x^4}} = +\infty$$

$(1+i)^4 = (\sqrt{2})^4(\cos(\frac{\pi}{4}) + i\sin(\frac{\pi}{4})) = -4$ De Moivre. □

Next proposition is analogous to propositions about the limits, continuity, and differentiability.

9.2 Properties of analytic functions

Proposition 9.8

If f, g are analytic functions on a domain Ω, then also $f + g$ and $f \cdot g$ are analytic on Ω. If $g(z) \neq 0$ on Ω, then $\frac{f}{g}$ is analytic on Ω. The composition $(g \circ f)$ of two analytic functions on Ω is an analytic function on Ω.

Proof. ***EXERCISE*** □

Example 9.9

Prove: if $f(z)$ is analytic and real on a domain Ω, then f is a constant function.

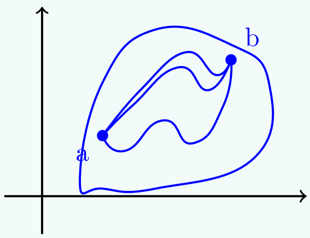
Proof. f is analytic and real on $\Omega \implies f(x, y) = u(x, y) + i \cdot 0$, namely $v(x, y) = 0$; CR equations hold for every $z \in \Omega$. f from analyticity on Ω: $\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} = 0$ and $\frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x} = 0 \implies u(x, y) = c \in \mathbb{R} \implies f(x, y) = c$, namely f is a constant function. □

Proposition 9.10

If f is analytic on a domain Ω and $f'(z) = 0$ for all $z \in \Omega$, then f is constant on Ω.

Remark 9.11. Prop 7.2 is false if Ω is not connected - f can take different constants (constant values) on each connected component of Ω.

Proof. The version of this proposition for real functions is proved using the Mean Value Theorem, that asserts that under appropriate continuity assumption for $f : (a, b) \to \mathbb{R}$ there exists $c \in (a, b)$ s.t. $f(b) - f(a) = f'(c)(b - a)$. So, if $f'(x) = 0$ for all $x \in (a, b)$, then $f(b) = f(a)$. The problem in the complex case is that there is no analogous idea of "between a and b" in $\mathbb{C}$, since there are many routes from a to b.

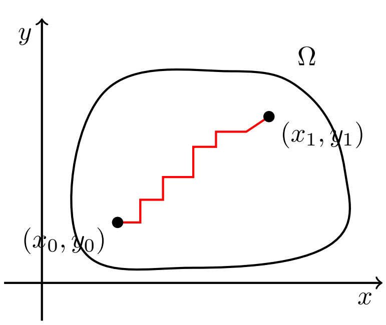


However, we can use the Mean Value Theorem for Re f and Im f as follows:

Since $f'(z) = 0$ for all $z \in \Omega$ by CR equations $0 = f'(z) = \frac{\partial u}{\partial x} + i\frac{\partial v}{\partial x}$ for all $z \in \Omega \implies \frac{\partial u}{\partial x} = \frac{\partial u}{\partial y} = \frac{\partial v}{\partial x} = \frac{\partial v}{\partial y} = 0$ for all $z \in \Omega$. ☐

Claim 9.12 — u is a constant function.

Proof. We have $u : \Omega \to \mathbb{R}, \frac{\partial u}{\partial x} = \frac{\partial u}{\partial y} = 0$ for all $(x, y) \in \Omega$. Pick $(x_0, y_0), (x_1, y_1) \in \Omega$. Ω is connected, thus by Def 6.2 any 2 points in Ω can be connected by a path that is entirely in Ω. We join (x_0, y_0) and (x_1, y_1) by a path comprised of pieces parallel to x-axis or y-axis [One needs to show that such path exists!] Since $\frac{\partial u}{\partial x} = 0$ by the Mean Value Theorem, u is constant on the horizontal pieces. In the same way, since $\frac{\partial u}{\partial y} = 0$, u is constant on vertical pieces. Thus, u is constant on Ω. By a similar argument we obtain that v is constant on Ω, therefore $f = u + iv$ is constant on Ω. ☐



Now we are going to study analyticity of multi-valued functions - here we will see the big difference between the complex and real functions.

10 Analyticity of multi-valued functions

10.1 Branch points and branches

We have seen that analytic function is always continuous, thus multi-valued functions are not analytic as is. The question is: can we choose a branch of such function that is single-valued and analytic?

Definition 10.1. We say that on a domain Ω there is a single-valued branch of multi-valued function if for every point of Ω we can choose one of the values of a function such that the resulting single-valued function is continuous in Ω.

In other words: to choose single-valued branch means to choose the domain of definition of a function such that when a point z moves along some closed continuous curve that is contained in the domain of definition and goes back to the starting point we get the same value of the function.

Let us demonstrate how we do that on the following example.

Example 10.2

Consider double-valued function $f(z) = \sqrt{z - z_0}$. $f(z)$ is defined for all $z \in \mathbb{C}$. Write: $z - z_0 = r(\cos\theta + i\sin\theta), r = |z - z_0|, \theta = \arg(z - z_0)$.

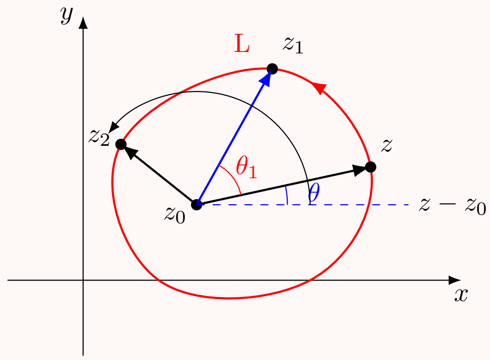
At every $z \neq z_0$, $f(z)$ attains 2 values ($\implies$ double-valued!).

$$w_0 = \sqrt{r}\left(\cos\frac{\theta}{2} + i\sin\frac{\theta}{2}\right)$$

$$w_1 = \sqrt{r}\left(\cos\left(\frac{\theta}{2} + \pi\right) + i\sin\left(\frac{\theta}{2} + \pi\right)\right) = -\sqrt{r}\left(\cos\frac{\theta}{2} + i\sin\frac{\theta}{2}\right) = -w_0$$

Let us follow the value w_0 when z is on the closed continuous curve L that encircles z_0 counterclockwise (anti-clockwise; the positive direction).

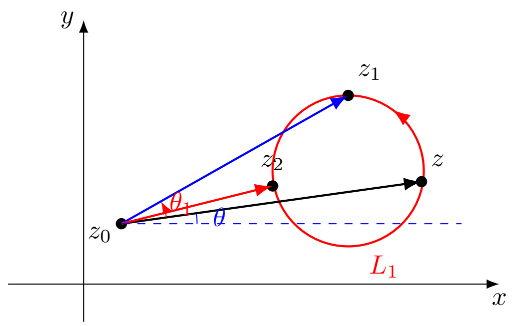
If z moves to z_1, the argument of $z_1 - z_0$ will be $\arg(z_1 - z_0) = \theta + \theta_1$.



If it continues to move along L, z will eventually return to the starting point: the modulus $r = |z - z_0|$ will be the same, but the argument will increase by 2π (since we completed the circle), and w_0 will attain the new value w_1.

$$\begin{aligned} w_{new} &= \sqrt{r}\left(\cos\left(\frac{\theta+2\pi}{2}\right) + i\sin\left(\frac{\theta+2\pi}{2}\right)\right) \\ &= \sqrt{r}\left(\cos\left(\frac{\theta}{2}+\pi\right) + i\sin\left(\frac{\theta}{2}+\pi\right)\right) = w_1 = -w_0 \end{aligned}$$

Let us check what happens to w_0 is we move along a closed continuous curve that does not encircle the point z_0.



From the picture above that the value of the modulus and of the argument after completing the full circle will return to its starting value - the same, with no change. This is independent of the curve L_1 (the only important thing that the curve does not encircle the point z_0).

Corollary 10.3

On a domain Ω that contains curves that encircle z_0, it is impossible to choose a continuous branch of the function $\sqrt{z - z_0}$.

Lecture 11+12: Continuation and Harmonic functions

We have studied the double-valued function $f(z) = \sqrt{z - z_0}$.

Reference, corollary 10.3.

For example: On $\frac{1}{2} < |z - z_0| < 2$, it is impossible to choose a single-valued branch of $\sqrt{z - z_0}$.

On $|z - z_0| > 1$, it is impossible to choose a single-valued branch of $\sqrt{z - z_0}$.

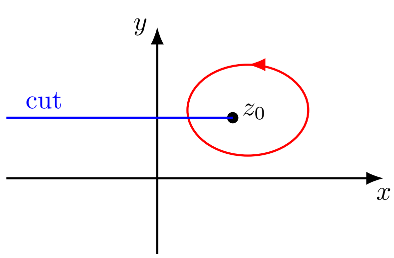
We have also seen that if the curve does not encircle z_0, then the value of $\sqrt{z - z_0}$ stays the same. So, we have another corollary.

Corollary 10.4
It is possible to choose a single-valued branch of the function $\sqrt{z - z_0}$ on a domain Ω if none of the closed continuous curves in Ω encircle z_0.

Intuitively, if we would be able to "cut" the plane (in some way) up to z_0, including z_0, then there will be no closed curve that encircles z_0. Indeed, the biggest domain on which none of the closed continuous curves encircle z_0 is the cutted plane, where we cut as follows.

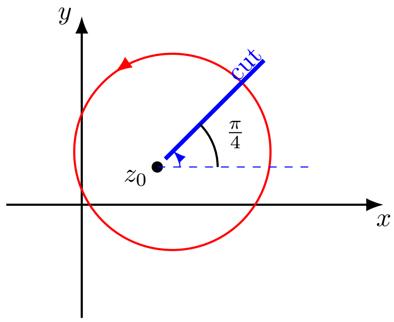
The "largest" such domain

- Cut-plane: $\mathbb{C}$ without the line $\{z = x + iy \mid x \leq x_0\}$ (including z_0).



- Cut-plane: two single-valued branches:

$$w_0 = \sqrt{r}(\cos\frac{\theta}{2} + i\sin\frac{\theta}{2}), \quad w_1 = -\sqrt{r}(\cos\frac{\theta}{2} + i\sin\frac{\theta}{2})$$



Note: We can cut along any ray starting at z_0.

The cut will prevent any closed curve from encircling z_0. In the case of the line in the picture above (ray creates an angle θ with the positive counterclockwise direction with the x-axis), we have:

For a ray with $\theta = \frac{\pi}{4}$, two continuous branches of $\sqrt{z - z_0}$:

$$w = \pm\sqrt{r}(\cos\frac{\theta}{2} + i\sin\frac{\theta}{2}), \quad \frac{\pi}{4} < \theta \leq \frac{5\pi}{4}$$

Definition 10.5. The point z_0 is called a *branch point* for a complex multi-valued function if the value of $f(z)$ does not return to its initial value as a closed curve around the point is traced (starting from some arbitrary point on the curve) in such a way that f varies continuously as the path is traced.

Important: What matters here is the **local** behavior of the function f near z_0. What may happen on paths that are some distance away from z_0 is not relevant. To be more precise:

The behavior must occur for all the curves that enclose the point and are sufficiently close to it.

In example 10.2 $z = z_0$ is the **Branch Point**.

Let us consider a more general case.

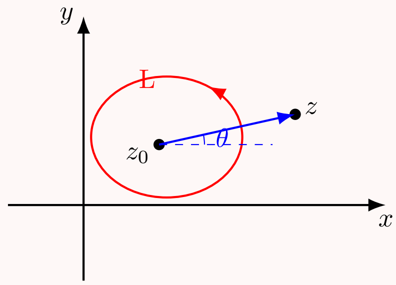
Example 10.6

Consider $w = \sqrt[n]{z - z_0}$. Take $z - z_0 = r(\cos\theta + i\sin\theta)$. Then:

$$w_k = \sqrt[n]{r}\left(\cos\frac{\theta + 2\pi k}{n} + i\sin\frac{\theta + 2\pi k}{n}\right), \quad k = 0, 1, 2, \ldots, n-1.$$

Fix $0 \leq k_0 \leq n-1$.

As before, we let z be on the closed continuous curve L that encloses z_0.



When z will complete the full circle (anticlockwise), the modulus r will return to the same value. The argument, however, will increase by 2π, and w will attain a new value.
After full circle around z_0:

$$w = \sqrt[n]{r}\left(\cos\frac{(\theta+2\pi)+2\pi k_0}{n} + i\sin\frac{(\theta+2\pi)+2\pi k_0}{n}\right).$$

$$= \sqrt[n]{r}\left(\cos\frac{\theta+2\pi k_0}{n} + i\sin\frac{\theta+2\pi k_0}{n}\right)\cdot\left(\cos\frac{2\pi}{n} + i\sin\frac{2\pi}{n}\right).$$

$$= w_{k_0}\left(\cos\frac{2\pi}{n} + i\sin\frac{2\pi}{n}\right) = w_{k_0+1}.$$

Namely, the new value of w is equal to the previous one times $\left(\cos\frac{2\pi}{n} + i\sin\frac{2\pi}{n}\right)$.

Check:

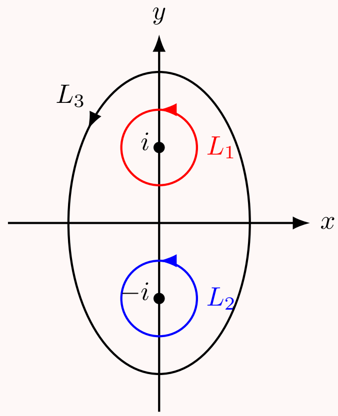
$$\left(\cos\frac{2\pi}{n} + i\sin\frac{2\pi}{n}\right)^n = 1.$$

This is independent of the curve that encloses z_0. If z runs on a curve that does not encircle z_0, then, as before, the value w_{k_0} will not change once z returns to the starting point.

Conclusion: If z is on the curve that encloses z_0 once (anticlockwise), then the value $\sqrt[n]{z-z_0} = w_{k_0}$ will change to the new value w_{k_0+1}. If z is on the curve that does not enclose z_0, then the value of the function stays the same. $\implies$ $z = z_0$ is the branch point of the function $f(z) = \sqrt[n]{z-z_0}$.

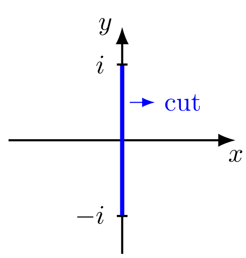
Example 10.7
Find the branch points of the function $f(z) = \sqrt{1+z^2}$.

Solution. $\sqrt{1+z^2} = \sqrt{z-i}\cdot\sqrt{z+i}$, therefore $z = \pm i$ are the branch points of $\sqrt{z-i}$, $\sqrt{z+i}$ $\implies$ branch points of $\sqrt{1+z^2}$. Let us see on the picture what happens and where to cut so we will be able to obtain a continuous single-valued branch of $f(z)$.



If z is on L_1 (around $z = i$): when z returns to the initial point $\sqrt{z+i}$ returns to the same value since L_1 does not enclose $z = -i$ while $\sqrt{z-i}$ attains new value - changes the sign of the previous one. Therefore, $f(z)$ also changes the sign. In the same way: if z is on L_2, then upon return to the initial point $\sqrt{z+i}$ will change the value - change the sign, $\sqrt{z-i}$ will return to the same value, and the function will change the sign. If z is on L_3 that encloses both $z_0 = i$ and $z_0 = -i$, then $\sqrt{z-i}$ and $\sqrt{z+i}$ will change the sign, but their product - the original function - will not! $\square$

$\Longrightarrow$ We can choose a continuous branch on a domain Ω if none of the closed curves in Ω enclose i and not $-i$ (or $-i$ and not i), but it can enclose both. Namely, if we cut the plane along the interval connecting i and $-i$, we will get a domain in which it is possible to choose a continuous branch.



Definition 10.8. Single-valued branch is called an analytic function on a domain Ω if it is differentiable at every $z \in \Omega$.

Example 10.9

Prove that $w = \sqrt[n]{z}$ is analytic on the domain $\Omega = \{z \in \mathbb{C} : 0 < \arg z < \frac{2\pi}{n}\}$.

Proof. We have already seen that on the cut-plane with the cut along any ray starting at $z_0 = 0$ (including $z_0 = 0$) every branch of this function is single-valued and continuous. Let us prove that it is differentiable and find the derivative.

$$\begin{aligned}\frac{f(z)-f(z_0)}{z-z_0} &= \frac{(f(z))^n-(f(z_0))^n}{z-z_0}\\ &= \frac{1}{(\sqrt[n]{z})^n-(\sqrt[n]{z_0})^n}\cdot\frac{(f(z))^{n-1}+(f(z))^{n-2}f(z_0)+\cdots+(f(z_0))^{n-1}}{z-z_0}\\ &= \frac{1}{(\sqrt[n]{z})^{n-1}+(\sqrt[n]{z})^{n-2}\sqrt[n]{z_0}+\cdots+(\sqrt[n]{z_0})^{n-1}}\end{aligned}$$

Now we pass to the limit as z tends to z_0 and obtain

$$\implies \lim_{z\to z_0}\frac{f(z)-f(z_0)}{z-z_0} = \frac{1}{n(\sqrt[n]{z_0})^{n-1}}$$

$\implies$ f is differentiable and

$$(\sqrt[n]{z})' = \frac{1}{n(\sqrt[n]{z})^{n-1}}.$$

□

In many problems in hydrodynamics, the elasticity theory and other fields there is often a need to recreate an analytic function from its real part. Now we will find the set of functions for which it is possible. Our subject is:

11 Harmonic functions.

11.1 Definitions and basic results

Definition 11.1. The real function $u(x,y)$ is called harmonic on a domain Ω if it is continuous on Ω with continuous partial derivatives of the first and second order

$$\{\frac{\partial u}{\partial x},\frac{\partial u}{\partial y},\frac{\partial^2 u}{\partial x^2},\frac{\partial^2 u}{\partial y^2},\frac{\partial^2 u}{\partial x\partial y},\frac{\partial^2 u}{\partial y\partial x}\},$$

and the following equation, called the Laplace equation, holds:

$$\frac{\partial^2 u}{\partial x^2}+\frac{\partial^2 u}{\partial y^2}=0$$

Example 11.2

The function $u(x,y)=x^2-y^2$ is harmonic for every $(x,y)\in\mathbb{R}^2$.

Solution. $u(x,y)$ is defined everywhere, and we have: $\frac{\partial u}{\partial x}=2x$ $\frac{\partial u}{\partial y}=-2y$ $\frac{\partial^2 u}{\partial x^2}=2$ $\frac{\partial^2 u}{\partial y^2}=-2$ $\frac{\partial^2 u}{\partial x\partial y}=\frac{\partial^2 u}{\partial y\partial x}=0$

All partial derivatives are continuous everywhere, and $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 2 + (-2) = 0$. $\square$

Example 11.3

The function $u(x, y) = e^x y$ is **not** harmonic function on $\mathbb{R}^2$.

Proof. $u(x, y)$ is defined everywhere on $\mathbb{R}^2$.

$$\frac{\partial u}{\partial x} = ye^{xy} \qquad \frac{\partial u}{\partial y} = xe^{xy}$$
$$\frac{\partial^2 u}{\partial x^2} = y^2 e^{xy} \qquad \frac{\partial^2 u}{\partial y^2} = x^2 e^{xy}$$
$$\frac{\partial^2 u}{\partial x \partial y} = e^{xy} + yxe^{xy} \qquad \frac{\partial^2 u}{\partial y \partial x} = e^{xy} + xye^{xy}$$

All partial derivatives are defined and continuous for all $(x, y) \in \mathbb{R}^2$. However $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = y^2 e^{xy} + x^2 e^{xy} = e^{xy}(x^2 + y^2) \neq 0$ except for $(0, 0) \implies u(x, y)$ is **not** harmonic function on $\mathbb{R}^2$. $\square$

Proposition 11.4

If the function $f(z) = u(x, y) + iv(x, y)$ is analytic on a domain Ω, then $u(x, y)$ and $v(x, y)$ are harmonic on Ω.

Proof. By CR equations $\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}$ and $\frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$ (since f is analytic). Let us differentiate the first equation with respect to x and the second with respect to y:

$$\frac{\partial^2 u}{\partial x^2} = \frac{\partial^2 v}{\partial y \partial x}, \quad \frac{\partial^2 u}{\partial y^2} = -\frac{\partial^2 v}{\partial x \partial y}$$

Since f is analytic, u and v are continuously differentiable, thus from theory of real functions, we know that $\frac{\partial^2 v}{\partial y \partial x} = \frac{\partial^2 v}{\partial x \partial y}$. Therefore, if we add the equations (*), we obtain:

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = \frac{\partial^2 v}{\partial y \partial x} - \frac{\partial^2 v}{\partial x \partial y} = \frac{\partial^2 v}{\partial x \partial y} - \frac{\partial^2 v}{\partial x \partial y} = 0$$

$\implies u$ is harmonic function on Ω.

If we differentiate now $\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}$ with respect to y and $\frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$ with respect to x, and use $\frac{\partial^2 u}{\partial x \partial y} = \frac{\partial^2 u}{\partial y \partial x}$ we will obtain that v is also harmonic function on Ω. $\square$

This proposition implies that in order to construct an analytic function we cannot choose the real and imaginary parts in an arbitrary way. The functions that we choose must be harmonic and, moreover, the CR equations must hold.

Definition 11.5. If $u(x, y)$ and $v(x, y)$ are harmonic functions on a domain Ω such that CR equations hold ($\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}, \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$), then the function $v(x, y)$ is called the *harmonic conjugate* of the function $u(x, y)$ on a domain Ω.

Proposition 11.6

The function $f(z) = u(x, y) + iv(x, y)$ is analytic in a domain $\Omega \iff v(x, y)$ is the harmonic conjugate of $u(x, y)$.

$\Longrightarrow$. Assume f is analytic in a domain Ω. By Prop 11.4 $u(x, y)$ and $v(x, y)$ are harmonic functions on Ω. Since f is analytic, CR equations hold, thus by Def 11.5 $v(x, y)$ is the harmonic conjugate of $u(x, y)$.

[$\Longleftarrow$] If $u(x, y)$ and $v(x, y)$ are harmonic functions on a domain Ω, then u and v are differentiable at all $z \in \Omega$. Since $v(x, y)$ is the harmonic conjugate of $u(x, y)$, by Def 11.5 - CR equations hold, thus Thm 6.1 8.13 asserts that f is analytic in a domain Ω. □

Thus, to conclude that we can reconstruct an analytic function from its real part we need to know whether it is always possible to find a harmonic conjugate to a given harmonic function. The answer is yes - it is given by the following proposition which we state without the proof.

Proposition 11.7

For any harmonic function $u(x, y)$ in a domain Ω there exists a function $v(x, y)$ in Ω which is the harmonic conjugate of $u(x, y)$.

The proof is by construction: one explicitly constructs the function $v(x, y)$. Since the proof involves somewhat complicated real analysis, we omit it.

Example 11.8

Find an analytic function $f(z) = u(x, y)+iv(x, y)$ such that $u(x, y) = \text{Re } f(z) = \arctan \frac{y}{x}$.

Solution. Let us compute the partial derivatives and then use CR equations: Recall: $(\arctan x)' = \frac{1}{1+x^2}$

$$\frac{\partial u}{\partial x} = \frac{-y}{x^2 + y^2} \qquad \frac{\partial u}{\partial y} = \frac{x}{x^2 + y^2} \implies \frac{\partial v}{\partial y} = \frac{-y}{x^2 + y^2}, \frac{\partial v}{\partial x} = \frac{-x}{x^2 + y^2}$$

CR equations

To obtain $v(x, y)$ we integrate $\frac{\partial v}{\partial x}$ with respect to x (or $\frac{\partial v}{\partial y}$ with respect to y).

$$v(x, y) = \int \frac{\partial v}{\partial x} dx = \int \frac{-x}{x^2 + y^2} dx = -\frac{1}{2} \ln(x^2 + y^2) + c(y)$$

Since the integral is with respect to x, the constant may depend on y (only!) To find $c(y)$ we differentiate $v(x,y)$ in (*) with respect to y and compare it to $\frac{\partial v}{\partial y}$ obtained from CR equations:

$$\frac{\partial v}{\partial y} = \frac{-y}{x^2+y^2} + c'(y) = \frac{-y}{x^2+y^2} \implies c'(y) = 0 \implies c = \text{constant} \in \mathbb{R}$$

$\implies v(x,y) = -\frac{1}{2}\ln(x^2+y^2) + c$ and $f(z) = \arctan\frac{y}{x} - \frac{i}{2}[\ln(x^2+y^2) + 2c]$.

$\square$

Week 5

Lecture 13: Sequences in $\mathbb{C}$

We introduce complex sequences and series, discuss their convergence and use them to construct complex power series. Then, we make several concrete connections with complex differentiable functions.

12 Complex sequences.

12.1 Sequence limits

Definition 12.1. A sequence $\{z_n\}_{n=0}^{\infty}, z_n \in \mathbb{C}$, has limit z (as $n \to \infty$) if for every $\epsilon > 0$ there exists $N = N(\epsilon) \in \mathbb{N}$ such that for all $n > N$, $|z_n - z| < \epsilon$. Notation: $\lim_{n\to\infty} z_n = z$.

If the limit does not exist (or ∞), we say that the sequence diverges.

In other words, a sequence of complex numbers $\{z_n\}$ converges to the complex number z iff at every neighborhood of z there are almost all but finite number of points from the sequence: $\lim_{n\to\infty} z_n = z \iff \forall\epsilon > 0 \quad \exists N(\epsilon) \in \mathbb{N}$ s.t. for every $n > N(\epsilon) \quad |z_n - z| < \epsilon$.

Claim 12.2 — If $\lim_{n\to\infty} z_n$ exists, then it is unique.

Proof. Assume $\lim_{n\to\infty} z_n = \tilde{z}_1$ and $\lim_{n\to\infty} z_n = \tilde{z}_2$. Then, for any $\epsilon > 0$ there exists $N_1 \in \mathbb{N}$ s.t. for every $n > N_1$, $|z_n - \tilde{z}_1| < \epsilon/2$. Also there exists $N_2 \in \mathbb{N}$ s.t. for every $n > N_2$, $|z_n - \tilde{z}_2| < \epsilon/2$. Choose $N = \max\{N_1, N_2\}$, then for every $n > N$, $|z_n - \tilde{z}_1| < \epsilon/2$ and $|z_n - \tilde{z}_2| < \epsilon/2$.

Thus $|\tilde{z}_1 - \tilde{z}_2| = |\tilde{z}_1 - z_n + z_n - \tilde{z}_2| \leq |\tilde{z}_1 - z_n| + |z_n - \tilde{z}_2| < \epsilon/2 + \epsilon/2 = \epsilon$. Since this is true for any $\epsilon > 0$ we conclude that $\tilde{z}_1 = \tilde{z}_2$. $\square$

Also, it is possible to show that the complex sequence $\{z_n\}$ converges to $z \iff$ the real sequence $\{|z_n - z|\}_{n=0}^{\infty}$ converges to 0 (**Check!**)

Example 12.3

Let $z_n = (\frac{i}{2})^n, n \in \mathbb{N}$. Then: $z_0 = 1, z_1 = \frac{i}{2}, z_2 = \frac{i^2}{4} = -\frac{1}{4}, z_3 = \frac{i^3}{8} = -\frac{i}{8}, z_4 = \frac{i^4}{16} = \frac{1}{16}, \ldots$

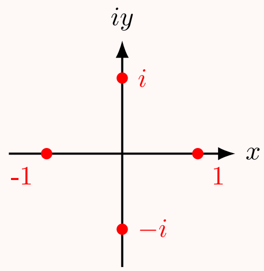
This sequence converges to 0: $\lim_{n\to\infty} z_n = 0$. Can be demonstrated by proving that $|z_n|$ decreases to 0 as n tends to infinity. (**Check:** By graphing the points z_n check that the sequence $\{z_n\}$ "spirals" in toward the origin (0) in the z-plane.

Example 12.4

Let $z_n = (2i)^n : z_0 = 1, z_1 = 2i, z_2 = -4, z_3 = -8i, \ldots$. This sequence diverges: one can show that $\{|z_n|\}_{n=0}^{\infty}$ is unbounded increasing sequence {namely, $|z_n|$ gets arbitrarily large}.

Example 12.5

Let $z_n = (i)^n : z_0 = 1, z_1 = i, z_2 = -1, z_3 = -i, z_4 = 1, z_5 = i, \ldots$. This is a bounded sequence ($|z_n| \leq 1$ for any n), but $\lim_{n\to\infty} z_n$ does not exist: there is no z such that almost all but finitely many points of the sequence are in a small neighborhood of some z:



z_n jumps at every step.

Important real sequence (limit): let $a_n = (1 + \frac{a}{n})^n$ for some a. Then $\lim_{n\to\infty} a_n = (1 + \frac{a}{n})^n = e^a$.

Example 12.6

$\lim_{n\to\infty}(1 + \frac{i}{n})^n = e^i, \quad \lim_{n\to\infty}(1 - \frac{i}{n})^n = \lim_{n\to\infty}(1 + \frac{-i}{n})^n = e^{-i} = \frac{1}{e^i}$.

Proposition 12.7

If the sequences $\{z_n\}, \{w_n\}$ of complex numbers converge, then so the sequences $\{z_n \pm w_n\}, \{z_n \cdot w_n\}$, and if $w_n \neq 0$ also $\{\frac{z_n}{w_n}\}$, and

$$\lim_{n\to\infty}(z_n \pm w_n) = \lim_{n\to\infty} z_n \pm \lim_{n\to\infty} w_n; \quad \lim_{n\to\infty} z_n w_n = \lim_{n\to\infty} z_n \lim_{n\to\infty} w_n.$$

If $\lim_{n\to\infty} w_n \neq 0$, then: $\lim_{n\to\infty} \frac{z_n}{w_n} = \frac{\lim_{n\to\infty} z_n}{\lim_{n\to\infty} w_n}$.

13 Complex (scalar) series.

13.1 Series convergence

Definition 13.1. The complex (scalar) series $\sum_{n=1}^{\infty} z_n$ converges to the value $s \in \mathbb{C}$ if the sequence of partial sums $\{S_N\}_{N=0}^{\infty}, S_N = z_0 + z_1 + \cdots + z_N$ converges to s. The value s is called the sum of the series. If the limit does not exist (or ∞) ($\lim_{N\to\infty} S_N$) then the series diverges. If the series converges to some value s, then s is unique.

Remark 13.2. Furthermore: if $r_N = S - S_N = \sum_{n=N+1}^{\infty} z_n$, then it is possible to show that $\sum_{n=0}^{\infty} z_n = S \iff \lim_{N\to\infty} r_N = 0$. Namely, the ***tail*** of the series converges to 0.

From Def (13.1) we immediately conclude:

Proposition 13.3

If $\sum_{n=0}^{\infty} z_n$ converges, then $\lim_{n\to\infty} z_n = 0$.

Proof. By Def (13.1): If $\sum_{n=0}^{\infty} z_n = S$, then $\lim_{N\to\infty} S_N = \lim_{N\to\infty} \sum_{n=0}^{N} z_n = S$. On the other hand: $S_N - S_{N-1} = (z_0 + z_1 + \cdots + z_N) - (z_0 + z_1 + \cdots + z_{N-1}) = z_N$ $\implies \lim_{N\to\infty} z_N = \lim_{N\to\infty}(S_N - S_{N-1}) = \lim_{N\to\infty} S_N - \lim_{N\to\infty} S_{N-1} = S - S = 0$ by Prop 12.7. s is unique. □

The contrapositive statement is often called the Test for Divergence.

Proposition 13.4 (Test for Divergence)

(Test for Divergence) Let $\{z_n\}_{n=0}^{\infty} \in \mathbb{C}$. If $\lim_{n\to\infty} z_n \neq 0$ (or does not exist), then the series $\sum_{n=0}^{\infty} z_n$ diverges.

Remark 13.5. The converse of Prop 13.4 is false: $\lim_{n\to\infty} \frac{1}{n} = 0$, but the series $\sum_{n=1}^{\infty} \frac{1}{n}$ diverges!

As expected there is an immediate relation between convergence or divergence of complex sequences and series to that of real ones.

Proposition 13.6

Let $\{z_n\}_{n=0}^{\infty}, z_n$ be the complex sequence and the complex series. Suppose that $z = x + iy$ and $S = u + iv$. Then:

1. $\lim_{n\to\infty} z_n = z \iff \lim_{n\to\infty} x_n = x$ and $\lim_{n\to\infty} y_n = y$ (for $z_n = x_n + iy_n$)
2. $\sum_{n=0}^{\infty} z_n = S \iff \sum_{n=0}^{\infty} x_n = u$ and $\sum_{n=0}^{\infty} y_n = v$

Proof. 1. follows directly from Def 12.1 and the application of triangle inequality: $|x_n - x|, |y_n - y| \leq |z_n - z| \leq |x_n - x| + |y_n - y|$

2. is proved by decomposing the complex sum into real and imaginary parts $\sum_{n=0}^{\infty} z_n = \sum_{n=0}^{\infty} x_n + i \sum_{n=0}^{\infty} y_n$ and a careful use of 1) [FINISH!]

□

Hint: If $\sum_{n=0}^{\infty} x_n = u, \sum_{n=0}^{\infty} y_n = v \implies$ obvious. Assume $\sum_{n=0}^{\infty} z_n = S$, define $X_N = \sum_{n=0}^{N} x_n, Y_N = \sum_{n=0}^{N} y_n$, apply 1) to $S_N = \sum_{n=0}^{N} z_n$.

In much of what follows, we can use Prop 11.3 and results in real sequences and series - these are left as an exercise. For example, Prop 10.1 and an analogous proposition for the series:

Proposition 13.7

Suppose that $\sum_{n=0}^{\infty} z_n = S, \sum_{n=0}^{\infty} w_n = t$ are two convergent series. Let $\alpha \in \mathbb{C}$. Then,

1. $\sum_{n=0}^{\infty}(z_n \pm w_n) = S \pm t$ (converges to $S \pm t$)
2. $\sum_{n=0}^{\infty} \alpha z_n = \alpha S$ (converges to αS)

Proof. (EXERCISE - exactly as in the real case - partial sums). □

Example 13.8

(Key example - Geometric series)

Suppose that $|z| < 1$. We claim that the series $\sum_{n=0}^{\infty} z^n$, called the Geometric series, converges.

Claim 13.9 — $\sum_{n=0}^{\infty} z^n$ converges (if $|z| < 1$) and $\sum_{n=0}^{\infty} z^n = \frac{1}{1-z}$.

Proof. We prove it using Def 11.1: we will show that the sequence of partial sums converges to $\frac{1}{1-z}$. Need to show: $\lim_{N\to\infty} S_N = \lim_{N\to\infty} \sum_{n=0}^{N} z^n = \frac{1}{1-z}$.

First, we obtain a closed formula for S_N as follows.

$$\begin{aligned} S_N &= 1 + z + z^2 + \cdots + z^N \\ z \cdot S_N &= z + z^2 + z^3 + \cdots + z^{N+1} \\ \implies S_N - zS_N &= S_N(1-z) = 1 - z^{N+1} \\ \implies S_N &= \frac{1 - z^{N+1}}{1-z} \end{aligned}$$

□

Now we use this formula to compute the limit

$$\left|\frac{1}{1-z} - S_N\right| = \left|\frac{1}{1-z} - \frac{1-z^{N+1}}{1-z}\right| = \left|\frac{z^{N+1}}{1-z}\right| = \frac{|z|^{N+1}}{|1-z|}$$

Since $|z| < 1 \implies \lim_{N\to\infty} |z|^N = 0$ (from what we know about real sequences). Therefore $\left|\frac{1}{1-z} - S_N\right| \xrightarrow{N\to\infty} 0 \implies \lim_{N\to\infty} S_N = \frac{1}{1-z}$.

Lecture 14+15: Absolute convergence?! WTH is that?

Now we introduce the notion of absolute convergence.

Definition 13.10. The series $\sum_{n=0}^{\infty} z_n$ converges absolutely if the series of the moduli of the terms $\sum_{n=0}^{\infty} |z_n|$ converges.

Remark 13.11. Note: (The Geometric series) For $|z| < 1$, $\sum_{n=0}^{\infty} |z|^n$ converges. Namely, the geometric series converges absolutely.

Proof. Follow exactly the same steps as in Ex 13.1, replace z by $|z|$. □

The definition of absolute convergence is subtle: Series can converge, but not absolutely: $\sum_{n=1}^{\infty} \frac{(-1)^n}{n}$ converges, but $\sum_{n=1}^{\infty} |\frac{(-1)^n}{n}| = \sum_{n=1}^{\infty} \frac{1}{n}$ diverges.

The absolute convergence *however* always implies convergence:

Proposition 13.12

If $\sum_{n=0}^{\infty} |z_n|$ converges, then $\sum_{n=0}^{\infty} z_n$ converges.

Proof. We have $|z_n| = \sqrt{x_n^2 + y_n^2} \geq |x_n|, |y_n|$, therefore using the Comparison test for positive (real) series we get that $\sum_{n=0}^{\infty} |x_n|$ and $\sum_{n=0}^{\infty} |y_n|$ converge $\implies$ (by analogous statement for real series) $\sum_{n=0}^{\infty} x_n$ and $\sum_{n=0}^{\infty} y_n$ converge. Thus, by Prop 13.6 2) $\sum_{n=0}^{\infty} z_n$ converges. □

Now we state 3 very useful tests for convergence of complex series.

Proposition 13.13

Let $\sum_{n=0}^{\infty} a_n, a_n \in \mathbb{C}$ be a complex series.

1. **Ratio-Test:** Consider the limit $\lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right| = \lim_{n\to\infty} \frac{|a_{n+1}|}{|a_n|} = L$
   - If the limit exists and $L < 1$, then the series $\sum_{n=0}^{\infty} a_n$ converges absolutely.

- If the limit exists and $L > 1$, then the series $\sum_{n=0}^{\infty} a_n$ diverges.
- If $L = 1$ or the limit does not exist, then the test is inconclusive {namely, there are examples of convergent and divergent series}.

2. **Root (Cauchy) Test:** Consider the limit $\lim_{n\to\infty} \sqrt[n]{|a_n|} = L$

- If the limit exists and $L < 1$, then the series $\sum_{n=0}^{\infty} a_n$ converges absolutely.
- If the limit exists and $L > 1$, then the series $\sum_{n=0}^{\infty} a_n$ diverges.
- If $L = 1$ or does not exist, then the test is inconclusive.

3. **Comparison Test:**

- If $0 \leq |a_n| < r_n$ and the positive real series $\sum_{n=0}^{\infty} r_n$ converges, then $\sum_{n=0}^{\infty} a_n$ converges absolutely.
- If $|a_n| > r_n \geq 0$ and $\sum_{n=0}^{\infty} r_n$ diverges, then $\sum_{n=0}^{\infty} a_n$ diverges.

Proof. Ratio Test: If $L > 1$, then the sequence $\{|a_n|\}$ is increasing (for sufficiently large n), therefore the series diverges by the Test for Divergence.

Suppose that $L < 1$. Choose a number r such that $L < r < 1$. Since $\lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right| = L$ there exists $N \in \mathbb{N}$ such that for all $n \geq N$: $\left|\frac{|a_{n+1}|}{|a_n|} - L\right| < \epsilon$. Set $\epsilon = a_N$. Then we have $|a_{N+1}| \leq r|a_N| = |a|r$, $|a_{N+2}| \leq r|a_{N+1}| < |a|r^2$ and, in general $|a_{N+k}| \leq |a|r^k$. Therefore for $n \geq N$ the terms of the series $\sum_{n=0}^{\infty} |a_n|$ are bounded by the terms of a convergent geometric series (since $0 \leq r < 1$), therefore $\sum_{n=0}^{\infty} |a_n|$ converges by the Comparison Test (for real series), namely, $\sum_{n=0}^{\infty} a_n$ converges absolutely.

Root Test and Comparison Test: EXERCISE! □

Example 13.14

Check the convergence of the following series:

1. $\sum_{n=0}^{\infty} (2i)^n$
2. $\sum_{n=0}^{\infty} \left(\frac{1+i}{2}\right)^n$
3. $\sum_{n=0}^{\infty} \frac{(4i)^n}{n!}$

Solution. 1. We use Comparison Test as follows: $a_n = (2i)^n \implies |a_n| = (|2i|)^n = |2|^n = 2^n > (\frac{3}{2})^n$. Since $\sum_{n=0}^{\infty} (\frac{3}{2})^n$ diverges (for example, by the Test for Divergence, since $\lim_{n\to\infty} (\frac{3}{2})^n = \infty \neq 0$) by Comparison Test $\sum_{n=0}^{\infty} (2i)^n$ diverges.

2. Let us apply the Root Test: $a_n = \left(\frac{1+i}{2}\right)^n \implies \lim_{n\to\infty} \sqrt[n]{|a_n|} = \lim_{n\to\infty} \sqrt[n]{\left|\frac{1+i}{2}\right|^n} = \lim_{n\to\infty} \frac{|1+i|}{2} = \frac{\sqrt{2}}{2} = \frac{1}{\sqrt{2}} < 1 \implies \sum_{n=0}^{\infty} \left(\frac{1+i}{2}\right)^n$ converges absolutely.

3. Let us apply the Ratio Test: $a_n = \frac{(4i)^n}{n!}$

$$\begin{aligned}\implies \lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right| = \lim_{n\to\infty} \left|\frac{\frac{(4i)^{n+1}}{(n+1)!}}{\frac{(4i)^n}{n!}}\right| &= \lim_{n\to\infty} \left|\frac{(4i)^{n+1}}{(n+1)!} \cdot \frac{n!}{(4i)^n}\right| \\ &= \lim_{n\to\infty} \frac{|4i|}{n+1} = \lim_{n\to\infty} \frac{4}{n+1} = 0 < 1\end{aligned}$$

$\implies$ converges absolutely.

□

14 Complex Power Series

14.1 Power series basics

Definition 14.1. The series $\sum_{n=0}^{\infty} a_n(z-z_0)^n$, $a_n \in \mathbb{C}, n \in \mathbb{N}$, is called a power series centered at $z = z_0$ (about $z = z_0$).

A power series is an example of complex series where the sequence z_n from Def 13.1 is of the special form $\{a_n(z-z_0)^n\}$ - involves variable z. Thus, the convergence or divergence of a power series may depend on the values of the variable z. If $z_0 = 0$, the power series are of the form $\sum_{n=0}^{\infty} a_n z^n$.

Remark 14.2. From Def 14.1 every power series $\sum_{n=0}^{\infty} a_n(z-z_0)^n$ converges for at least one point $z = z_0$.

Next theorem tells us that convergence at one point guarantees convergence on some small disc around z_0. We prove it for the case $z_0 = 0$ (for simplicity of notation), the general case $z_0 \neq 0$ is proved in exactly the same way.

Theorem 14.3 (Abel)
If $\sum_{n=0}^{\infty} a_n z^n$ converges at $z_0 \neq 0$, then it converges absolutely for any z with $|z| < |z_0|$. Namely, if the series converges at a point z_0, then it converges absolutely at every interior point of the disc centered at 0 of radius $|z_0|$.

Proof. Since $\sum_{n=0}^{\infty} a_n z_0^n$ converges by (Prop 13.3) $\lim_{n\to\infty} a_n z_0^n = 0$, thus $\{a_n z_0^n\}$ is a bounded sequence: there exists $M \in \mathbb{N}$ such that for every $n \in \mathbb{N}$ $|a_n z_0^n| < M$. Write the modulus of the general term of the series as: $|a_n z^n| = |a_n z_0^n \cdot \frac{z^n}{z_0^n}| = |a_n z_0^n| \left|\frac{z}{z_0}\right|^n < M \left|\frac{z}{z_0}\right|^n \implies$ since $|z| < |z_0|$, namely $\left|\frac{z}{z_0}\right| < 1$, the general term of the positive series $\sum_{n=0}^{\infty} |a_n z^n|$ is smaller than the one of the positive geometric series $\sum_{n=0}^{\infty} M \left|\frac{z}{z_0}\right|^n \implies$ by Comparison Test ($\sum_{n=0}^{\infty} |a_n z^n|$) converges $\implies \sum_{n=0}^{\infty} a_n z^n$ converges absolutely. □

Corollary 14.4

If $\sum_{n=0}^{\infty} a_n z^n$ converges at $z = z_1$, then it diverges for every z with $|z| > |z_1|$.

Definition 14.5. If the series $\sum_{n=0}^{\infty} a_n z^n$ converges for all $z \in U \subseteq \mathbb{C}$, then its sum defines a function on U.

Example 14.6 (13.8(continued))

$\sum_{n=0}^{\infty} z^n$ - power series centered at $z_0 = 0$, $a_n = 1$ for all n. Converges on the open unit disc $U = \{z \in \mathbb{C} : |z| < 1\}$ (converges absolutely!), and equal to the function $f(z) = \frac{1}{1-z}$ on U.

Next proposition tells us where the power series converges. The proof is omitted.

Proposition 14.7

Given a power series $\sum_{n=0}^{\infty} a_n (z - z_0)^n$ there exists $0 \leq R \leq \infty$ {a real non-negative number or else infinity} such that if $|z - z_0| < R$, then $\sum_{n=0}^{\infty} a_n (z - z_0)^n$ converges absolutely and if $|z - z_0| > R$, then $\sum_{n=0}^{\infty} a_n (z - z_0)^n$ diverges.

The value R is called the radius of convergence of power series.

A very useful formulae for computing the radius of convergence of power series:

- $R = \lim_{n\to\infty} \left|\frac{a_n}{a_{n+1}}\right|$ {for $\sum_{n=0}^{\infty} a_n (z - z_0)^n$}
- Cauchy-Hadamard formula: $R = \lim_{n\to\infty} \frac{1}{\sqrt[n]{|a_n|}}$

Example 14.8

Find the radius of convergence of $\sum_{n=0}^{\infty} (n + i) z^n$.

Solution. $a_n = n + i \implies |a_n| = \sqrt{n^2 + 1}$.

Using Cauchy-Hadamard formula and that $\lim_{n\to\infty} \sqrt[n]{n} = 1$ we get:

$$R = \lim_{n\to\infty} \frac{1}{\sqrt[n]{\sqrt{n^2+1}}} = 1.$$

Namely, the series converges absolutely for $|z| < 1$. $\square$

Remark 14.9. The power series with the radius of convergence R on the circle $|z - z_0| = R$ (or $|z| = R$ if $z_0 = 0$) may converge or diverge or converge at some points and diverge at others.

Example 14.10

1. The geometric series $\sum_{n=0}^{\infty} z^n$ - the radius of convergence $R = 1$ and the series diverges at any point on the circle $|z| = 1$ (by the Test for Divergence).
2. $\sum_{n=0}^{\infty} \frac{z^n}{n^2}$: The radius of convergence $R = 1$ [**Check!**]

   and the series converges (absolutely) for any $|z| \leq 1$.
3. One can prove: the series $1 + z + \frac{z^2}{2} + \frac{z^3}{3} + \cdots + \frac{z^n}{n} + \ldots$ with the radius of convergence $R = 1$, diverges at $z = 1$ and converges at all other points $|z| = 1$ (not absolutely!).

Definition 14.11. The disc of convergence of the power series $\sum_{n=0}^{\infty} a_n(z - z_0)^n$ is given by $D = \{z \in \mathbb{C} : |z - z_0| < R\}$ and we write $f(z)$ for the sum of the series on this disc.

Proposition 14.12

Suppose the power series $\sum_{n=0}^{\infty} a_n(z - z_0)^n$ has disc of convergence D. Then,

1. $f(z) = \sum_{n=0}^{\infty} a_n(z - z_0)^n$ is analytic on D {differentiable at every $z \in D$}
2. For all $z \in D$ the series $\sum_{n=0}^{\infty} n a_n(z - z_0)^{n-1}$ is absolutely convergent and has sum $f'(z)$ (the derivative of f at z).

Note: From Prop 14.12: $f(z) = \sum_{n=0}^{\infty} a_n(z - z_0)^n$ is differentiable arbitrarily many times at any z in the disc of convergence of the series (namely, we can define its higher derivatives).

Definition 14.13. For any $z \in \mathbb{C}$ we define the exponential function as the sum of the following series

$$e^z = 1 + z + \frac{z^2}{2!} + \frac{z^3}{3!} + \cdots = \sum_{n=0}^{\infty} \frac{z^n}{n!}$$

If z is real, we get the usual expansion of e^x to series.

Claim 14.14 — Claim: e^z is well-defined for all $z \in \mathbb{C}$, namely $R = \infty$.

Proof. By Ratio Test:

$$\lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| = \lim_{n\to\infty}\left|\frac{\frac{z^{n+1}}{(n+1)!}}{\frac{z^n}{n!}}\right| = \lim_{n\to\infty}\left|\frac{z}{n+1}\right| = 0 < 1 \quad \text{for all } z \in \mathbb{C}$$

$\implies \sum_{n=0}^{\infty} \frac{z^n}{n!}$ converges absolutely for all $z \in \mathbb{C}$. □

Proposition 14.15

e^z is entire and $(e^z)' = e^z$.

Proof. By Prop 14.12 1) e^z is analytic for every $z \in \mathbb{C} \implies$ entire (since $R = \infty$). By Prop 14.12 2): $(e^z)' = \sum_{n=1}^{\infty} \frac{nz^{n-1}}{n!} = \sum_{n=1}^{\infty} \frac{z^{n-1}}{(n-1)!} = \sum_{n=0}^{\infty} \frac{z^n}{n!} = e^z$. □

Proposition 14.16

For any $z_1, z_2 \in \mathbb{C} : e^{z_1} \cdot e^{z_2} = e^{z_1+z_2}$.

Proof. By Def 14.13: $e^{z_1} = \sum_{n=0}^{\infty} \frac{z_1^n}{n!}, \quad e^{z_2} = \sum_{n=0}^{\infty} \frac{z_2^n}{n!}$.

These are absolutely convergent series $\implies$ we can multiply them: $e^{z_1+z_2} = 1 + (z_1 + z_2) + \frac{1}{2!}(z_1 + z_2)^2 + \cdots + \frac{1}{n!}(z_1 + z_2)^n + \cdots = 1 + (z_1 + z_2) + \left(\frac{z_1^2}{2!} + z_1 z_2 + \frac{z_2^2}{2!}\right) + \cdots + \left(\frac{z_1^n}{n!} + \frac{z_1^{n-1}}{(n-1)!} z_2 + \cdots + \frac{z_2^n}{n!}\right) + \cdots = e^{z_1} e^{z_2}$ □

Now we can define trigonometric functions as power series:

Definition 14.17. For any $z \in \mathbb{C}, |z| < \infty$

$$\sin z = \operatorname{Im} e^{iz} = z - \frac{z^3}{3!} + \frac{z^5}{5!} - \cdots = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} z^{2n+1}$$

$$\cos z = \operatorname{Re} e^{iz} = 1 - \frac{z^2}{2!} + \frac{z^4}{4!} - \cdots = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!} z^{2n}$$

For real z this definition coincide with Taylor series expansion of $\sin x$ and $\cos x$ about $x = 0$.